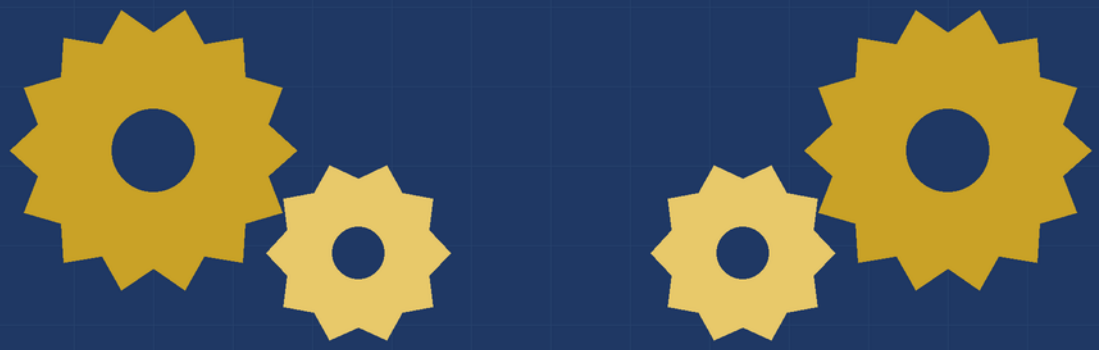
MIND AI x ONTYPOLOGY



AI Mechanical Design & PLM

the complete accurate-output method

with Ontypology (Space Type × Time Type) as the theoretical basis

base platform: Mind AI

every design output can state where it came from

Type = Type (not Form)

SpecType	PartType	AssemblyType	InspectType	ChangeType	HistoryType
Spec	Part	Assembly	Inspect	Change	History

accurate-output method · complete solution · pastel illustrated edition

2026.10

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Before we begin: three conventions

This is a book about how AI can make mechanical design and PLM document output accurate. It does not retell theory or exhibit system code; it applies a set of cognitive tools directly on the mechanical R&D floor, so that in all six outputs - specification, part, assembly, inspection, change, history - every number can state where it came from.

So that every later chapter runs on the same track, three conventions are set first.

> **Convention one: the English word for Type is Type**
>
> The word "Type" in this book is always rendered Type in English, never Form.
> Hence SpecType, PartType, AssemblyType and so on; Type-Driven Design is "design driven by Types".

> **Convention two: the theory is used, not taught**
>
> Ontypology (the Space Type and the Time Type) is the theoretical basis of this book, but this book does not restate its text. This book only uses its tools: the three brackets [ ] { } ( ), the intersection criterion ∩(T), the identity anchor id(C), the three levels of remove, the four-step framework, the three-layer judgement and the move tray.
> For the derivation behind each tool, go back to the original theoretical work; this book is about using the tools in the right place.

> **Convention three: the platform is cited, not restated**
>
> The underlying platform is Mind AI: it provides a unified knowledge model, a reasoning engine, memory and self-evolution.
> This book does not restate the platform, only cites its place in the division of labour where needed - the engineer talks to Mind AI, while criteria and evidence stay in the Type Library.

The three conventions in one sentence: Type is Type, use the theory as a tool, use the platform as the base.

Task Map: How to Use This Book

The book's six chapters fall into four stretches: first meet the brain (Chapter 1), then meet its output (Chapter 2), then see how accuracy is guaranteed (Chapter 3) and how it is produced (Chapters 4 and 5), and finally turn the mirror on yourself (Chapter 6 and the review). The table below is the task map - chapters on the left, what you take away from each on the right.

Table 0-1 Task map: what each chapter gives you

Chapter	Topic	What you get
Chapter 1	what this brain is	the concept of the reasonable product replica, the mapping of the two Types onto the R&D process, the platform division of labour, and the four sources of accuracy
Chapter 2	the six output Types	the definitions, fields and examples of the six Types (spec, part, assembly, inspection, change, history); the BOM criterion and the three levels of remove
Chapter 3	the four gates of accuracy	∩(T) criterion funnel, deductive derivation chain, engineering validation, Type evidence chain
Chapter 4	the six-step method of working	the four-step framework, the six-step flow, the three questions of Type logic, and the selection and combination of moves
Chapter 5	prompts and examples	The full text of the S1-S7 prompts, plus the complete planetary gearbox output from requirement to history
Chapter 6 and the review	self-check	misuse comparison, the ten questions, maturity L1-L4, the three closing questions

In Table 0-1 the left column deliberately uses half-width "Chapter N" numbers, kept separate from the chapter titles in the contents, so that machines and people can each look things up.

Three ways to read. The engineer in a hurry: go straight to the six-Type field tables in Chapter 2 and the prompts in Chapter 5, and put them to work. The manager leading a team: read Chapters 1, 3 and 6 for the acceptance standards and the team maturity ladder. The reader who wants a full rollout: read in order and walk Chapter 5's example through your own case.

Introduction

D-1 Two Languages, One Machine

Mechanical engineering has its own language: specification, tolerance, fit, BOM, ECN, revision. Ontypology has its own too: Type, property, composition, container, criterion, move. The two languages describe the same machine - the difference is that engineering language tells you which fields to fill, while Type language tells you how those fields relate.

Table D-1 The Two Languages Compared

engineering language	The language of Ontypology	the corresponding output Type
specification, requirement clauses	Property [ ] and property values, the intersection criterion ∩(T)	Spec Type
part drawing, material list	Unit Type, BOM set { }	Part Type (PartType)
assembly drawing, BOM	composition structure, Spatial Container ( )	Assembly Type (AssemblyType)
inspection report, measurement data	criterion acceptance, three-layer judgement	Inspection Type (InspectType)
engineering change notice ECN	three levels of remove, dependency-chain impact analysis	Change Type (ChangeType)
revision history, configuration management	identity anchor id(C), Time Type evolution	History Type (HistoryType)

Table D-1 The same planetary gearbox: on the left what the engineer writes every day, on the right the structure AI can reason about.

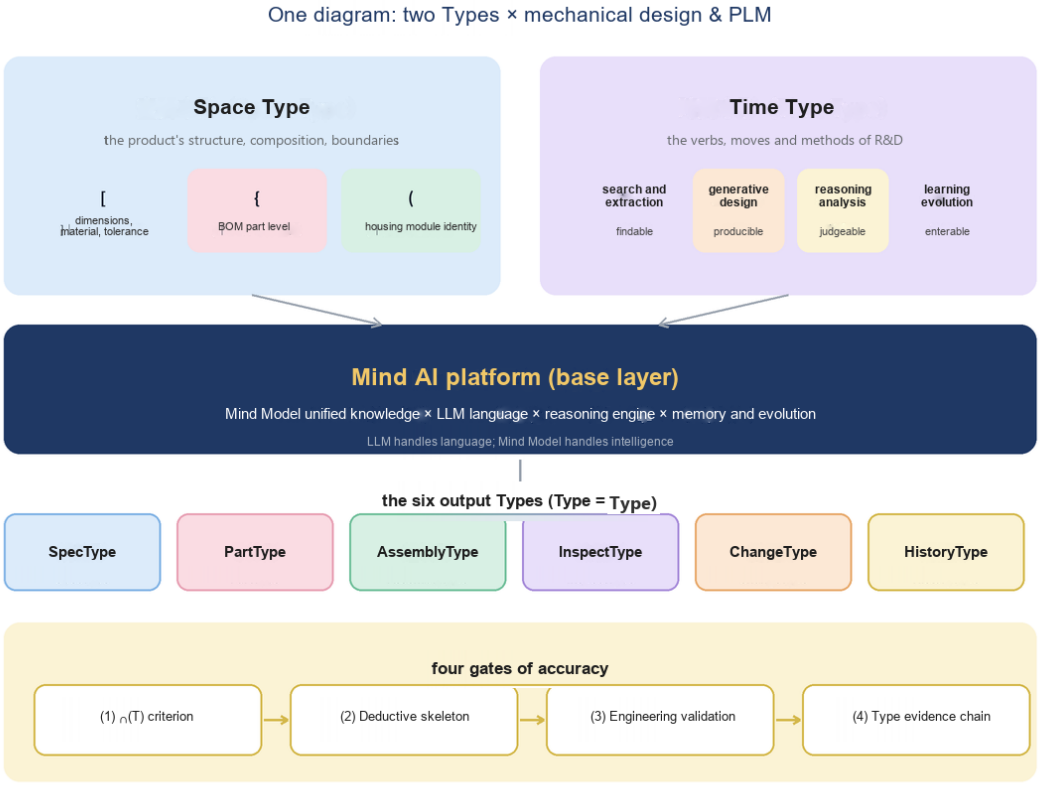


Figure D-1 Two Types × mechanical design and PLM panorama: space raises the skeleton, time supplies the moves, the platform provides the base, six Types form the output

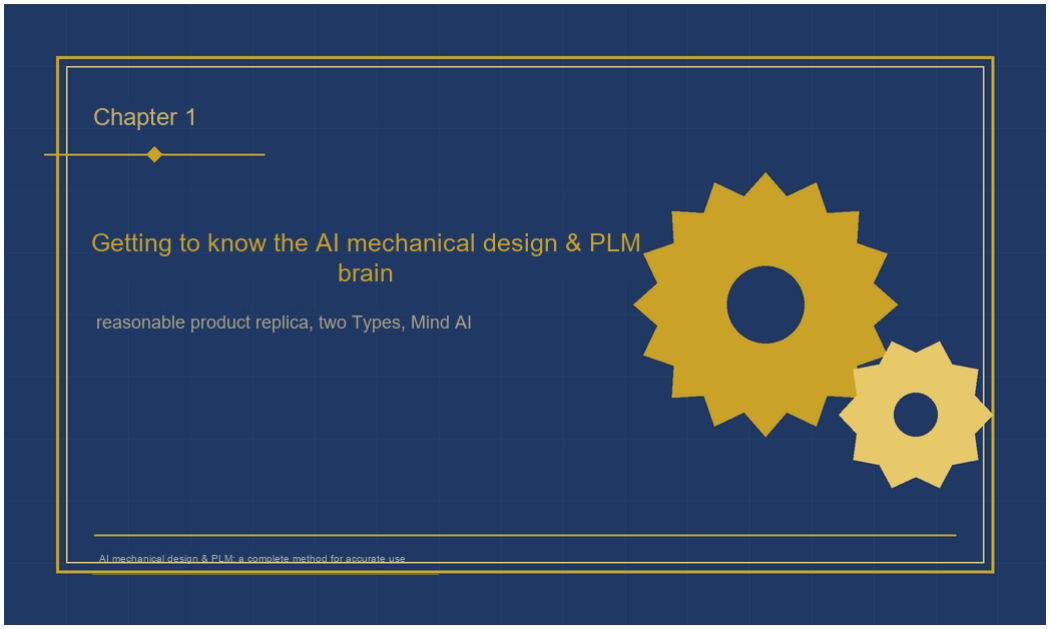
D-2 The Panorama and Three Ways to Read

Figure D-1 is the whole book compressed. The upper half is the two Types: the Space Type raises the product structure with three axes and three brackets, the Time Type turns R&D actions into repeatable methods with the move tray. In the middle is the Mind AI platform - the LLM supplies language ability, the Mind Model supplies reasoning and memory, and the two merge. The lower half is the output chain strung from the six output Types, and the four accuracy gates that guard them.

Along this diagram you can enter the book three ways. Read along the output chain: from SpecType through to HistoryType, for those racing to finish documents. Read along accuracy: check the four gates against the case in hand, for those doing acceptance. Read along the growth path: climb from L1 dialogue output to L4 self-evolution, for those planning team capability.

> **Chapter focus**
>
> Two languages describe the same machine: engineering language gives the fields, Type language gives the relations.
> How to remember the panorama: the two Types on top, the platform in the middle, the six Types and the four accuracy gates below.
> Pick one way in first; when you have finished the book, come back and read it another way.

Chapter 1 Getting to Know the AI Mechanical Design and PLM Brain



Let the protagonist take the stage: an AI mechanical design and PLM brain. It is not a CAD assistant that chats, nor a document generator - it is a brain that remembers what the product is, reasons out why the design is so, and calculates who is affected after a change. This chapter uses four sections to set out its position: the product replica, where the two Types land, the platform division of labour, and where the four words "accurate output" actually come from.

1.1 A Reasonable Product Replica: Type + Number

Every mechanical case has a "replica" in the computer: BOM, drawings, specification, change records. But the traditional replica is a pile of files - people can read them, AI cannot read the relations. A reasonable product replica requires everything in it to be a Type: parts are Types, assemblies are Types, specifications are Types, changes are Types. A Type has a name, boundaries, composition and criteria, so AI can question it and get meaningful answers.

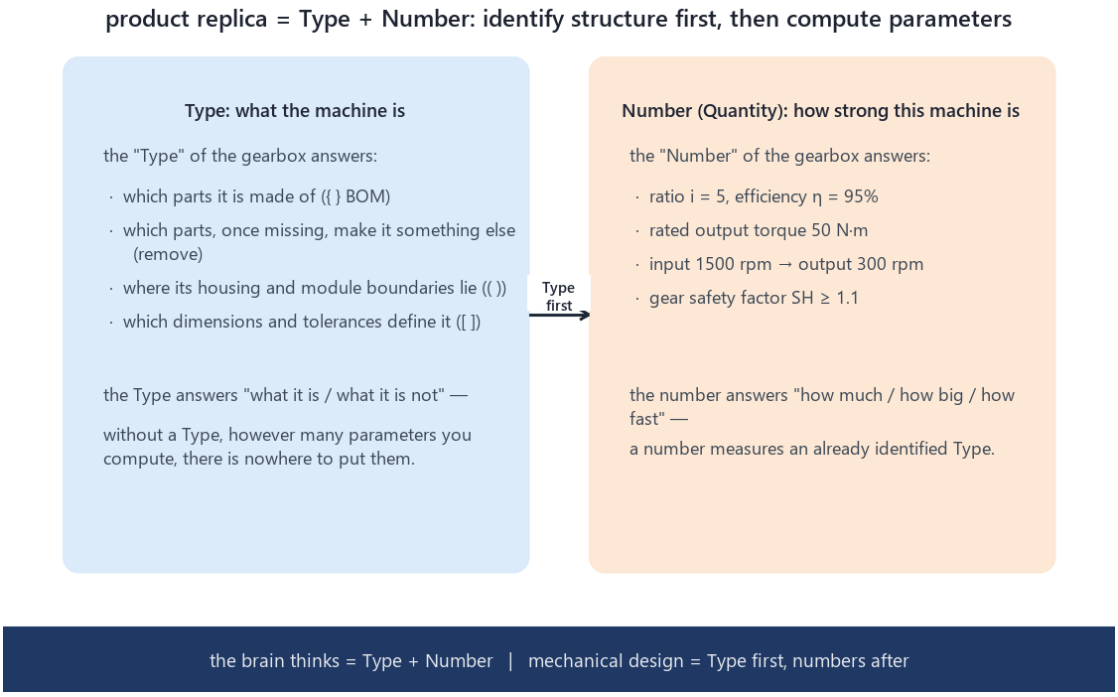


Figure 1-1 Product replica = Type + Number: identify the structure first, then compute the parameters

Take the running case of this book, the single-stage planetary gearbox. The Type "planetary gearbox" answers what it is: it consists of a sun gear, planet gears, a ring gear, a planet carrier and a housing, and without the sun gear it cannot transmit motion. The numbers i = 5, η = 95% and rated output torque 50 N·m answer how strong it is. Numbers cannot exist apart from the Type: without the Type "planetary gear train", the number "27 teeth" means nothing. First the Type, then the number - that is the first habit of accurate output.

Table 1-1 The four scenes of "inaccurate output" and the Type prescription

On-site symptom	the real cause	the Type prescription
the same part does not match across documents	no Types established, names drift	establish the PartType and anchor its identity with id(C)
the specification is written but cannot be accepted	no quantified criteria	the SpecType has built-in ∩(T) acceptance clauses
documents left unrevised after a change	no dependency chain	AssemblyType BOM + three-level remove impact analysis
AI output looks reasonable but nobody dares use it	the number has no origin	Type evidence chain: every number is tagged with its source

Table 1-1 The four symptoms share one cause: Type decoupled from number. The next five chapters fit these four prescriptions back one by one.

1.2 The Two Types Land in Mechanical R&D

The Space Type handles "what the product looks like": the X axis property [ ] circles out specifications and tolerances, the Y axis composition { } opens up the BOM levels, and the Z axis container ( ) draws the boundaries of housing, modules and installation space. The Time Type handles "how R&D moves": analysis, design, reasoning, refactoring, self-evolution - each verb is a move that can be applied again and again.

Table 1-2 The two Types × where they land in mechanical R&D

the Ontypology tools	where it lands in mechanical R&D	practical example
[ ] Property (X axis)	specification clauses, parameters, tolerances	module 1.5, pressure angle 20°, accuracy ISO 1328 grade 6
{ } Composition (Y axis)	BOM levels, part composition	L0 machine → L1 gear train → L2 sun gear
( ) Container (Z axis)	housing, module, storage-location boundaries	the planetary train is enclosed in the housing; lubricant must not cross the boundary
Time Type - moves	the verbs of the R&D process	requirement analysis, selection design, strength reasoning, ECN refactoring
id(C) identity anchor	configuration management	revisions never break: A → B is still the same machine
Three levels of remove	change impact analysis	removing the sun gear = fatal; changing the nameplate = imperceptible

Table 1-2 The three space axes raise the skeleton, the time moves fix the actions, and id(C) and remove manage change.

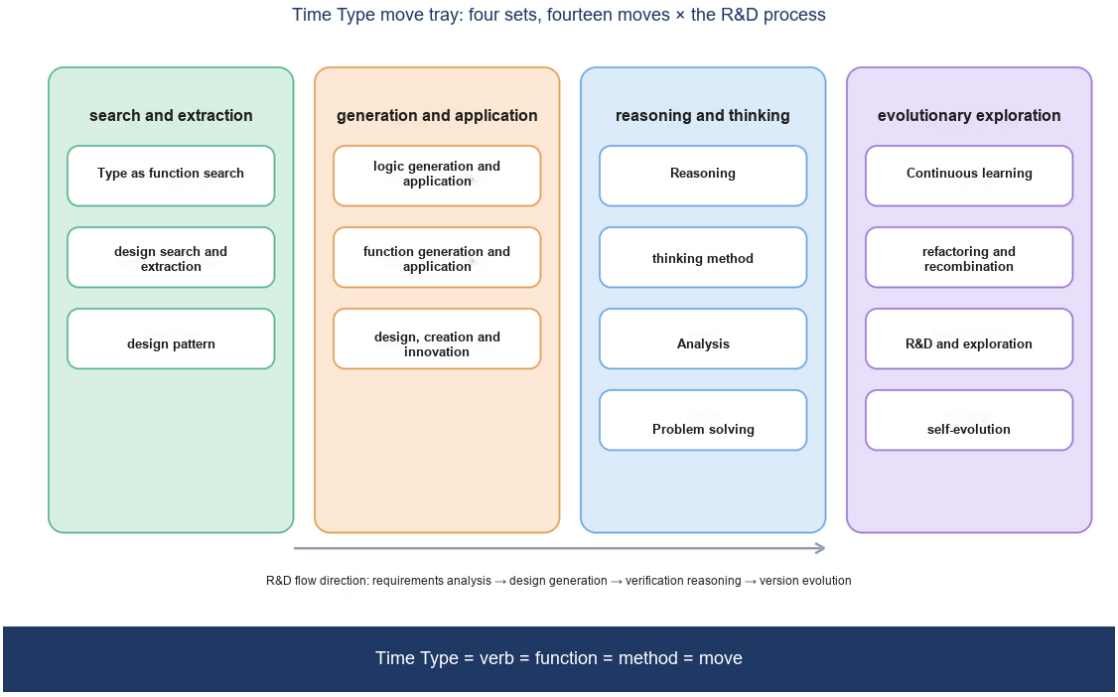


Figure 1-2 The Time Type move tray: four sets and fourteen moves mapped to the verbs of the R&D process

The full list of the fourteen moves and their mechanical R&D mapping is in Table A-2 of Appendix A. For now remember one thing: a move is a Time Type - it equals a method, equals a function - with an input (situation), an action (method) and an output (deliverable), so the same move can be used again on the next case.

1.3 The Division of Labour between Mind AI and the LLM

Beneath the brain is the Mind AI platform. It is not a single model but an architecture: the LLM provides language understanding and generation, the Mind Model provides unified knowledge, deep reasoning, long-term memory and self-evolution, and a reasoning engine and planning system schedule the two. In one sentence: the LLM handles language, the Mind Model handles intelligence.

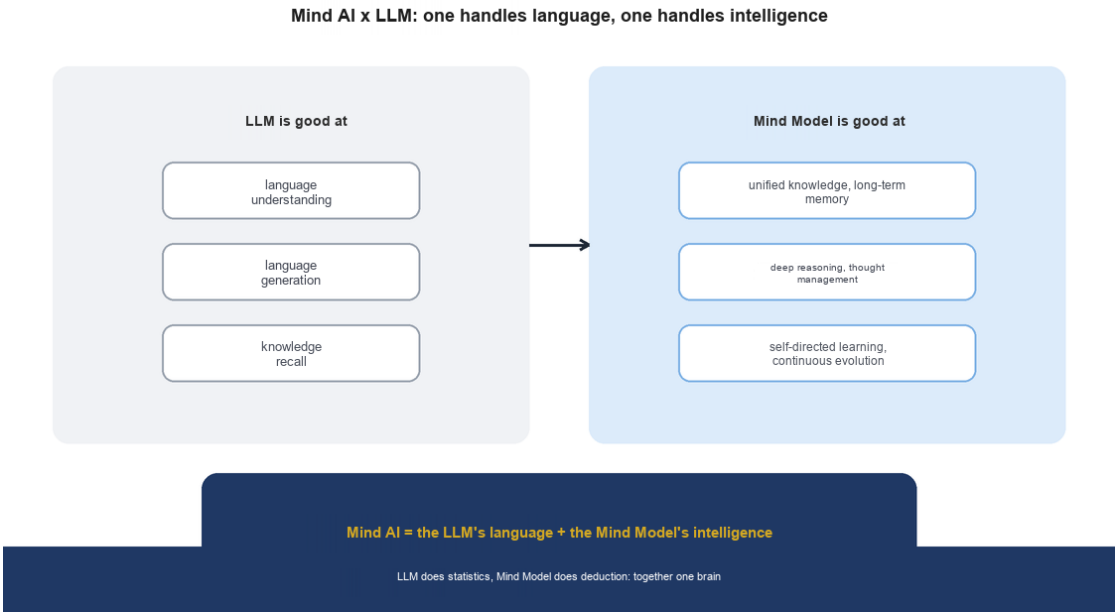


Figure 1-3 Mind AI × LLM division of labour: language to the LLM, intelligence to the Mind Model

Table 1-3 Platform division of labour: who is responsible for what

component	the intelligence of responsibility	what it looks like in mechanical design and PLM
LLM	language understanding and generation	turn the engineer's conversation into a SpecType and report documents
Mind Model	unified knowledge and memory	Type Library: PartTypes, the move library and the criteria library accumulate over time
reasoning engine	deduction and causal reasoning	from i = 5, Zr/Zs = 4 follows, step by step checkable
planning system	task decomposition and scheduling	break a case into the seven-stage S1-S7 pipeline
learning system	continuous learning and evolution	misuse cases feed back to the Type Library and are screened automatically next time

Table 1-3 The engineer talks to Mind AI; what really remembers the product is the Type Library.

This division of labour matters directly for accuracy: the text an LLM generates carries a statistical blur by nature, while the criteria in the Type Library are definite. Let language stay language and criteria stay criteria, and only then does AI output deserve a stamp.

1.4 Where "Accurate Output" Comes From

"Accurate output" is not a slogan but the composite of four gates. The first is criteria: every Type carries an ∩(T) intersection criterion, and only output that passes counts. The second is derivation: conclusions are not copied, they are derived step by step along the deductive skeleton. The third is validation: the numbers must pass engineering validation - tolerance chain, strength, performance. The fourth is traceability: the Type evidence chain lets every number be traced back to its source.

Table 1-4 The four gates of accuracy (preview)

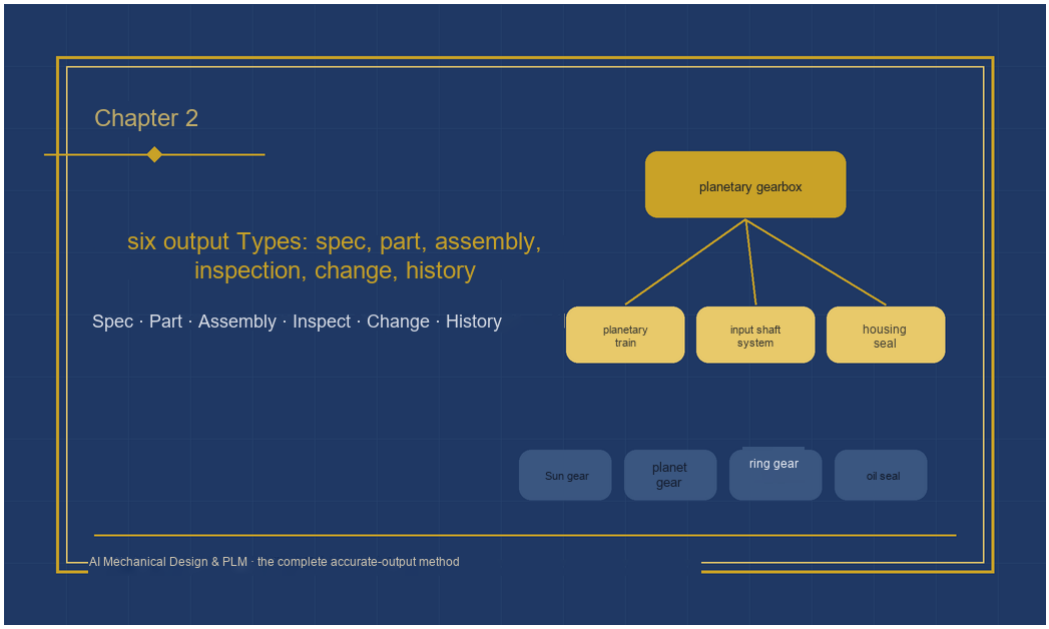
Layer	what it guards	One sentence
First layer	∩(T)	if the intersection is not satisfied, it does not enter the Type
Second layer	Deductive skeleton	conclusions are derived, not copied
Third layer	engineering validation	computing the number is not enough; it counts only when validation passes

Layer	what it guards	One sentence
Fourth layer	Type evidence chain	every output traces back to a Type and a criterion

Table 1-4 Chapter 3 takes these four gates apart one by one; this chapter only plants the landmarks.

> **Chapter focus**
>
> A reasonable product replica = everything is a Type; Type first, number second.
> The two Types: the three space axes raise the structure, the time moves fix the method.
> The LLM handles language, the Mind Model handles intelligence - accuracy comes from criteria, not from fluency.
> Accurate output = criteria + derivation + validation + traceability; none of the four gates may be missing.

Chapter 2 The Six Output Types: Spec, Part, Assembly, Inspection, Change, History



Mechanical design and PLM produce many documents but only six kinds of output: specification, part, assembly, inspection, change, history. Set each of the six up as a Type and you have six output Types - remembered in this book as "spec, part, assembly, inspection, change, history". The first three design the product; the last three govern it. Every Type has definition fields (what it records), criteria (what counts as passing), and upstream and downstream neighbours (whom it connects to).

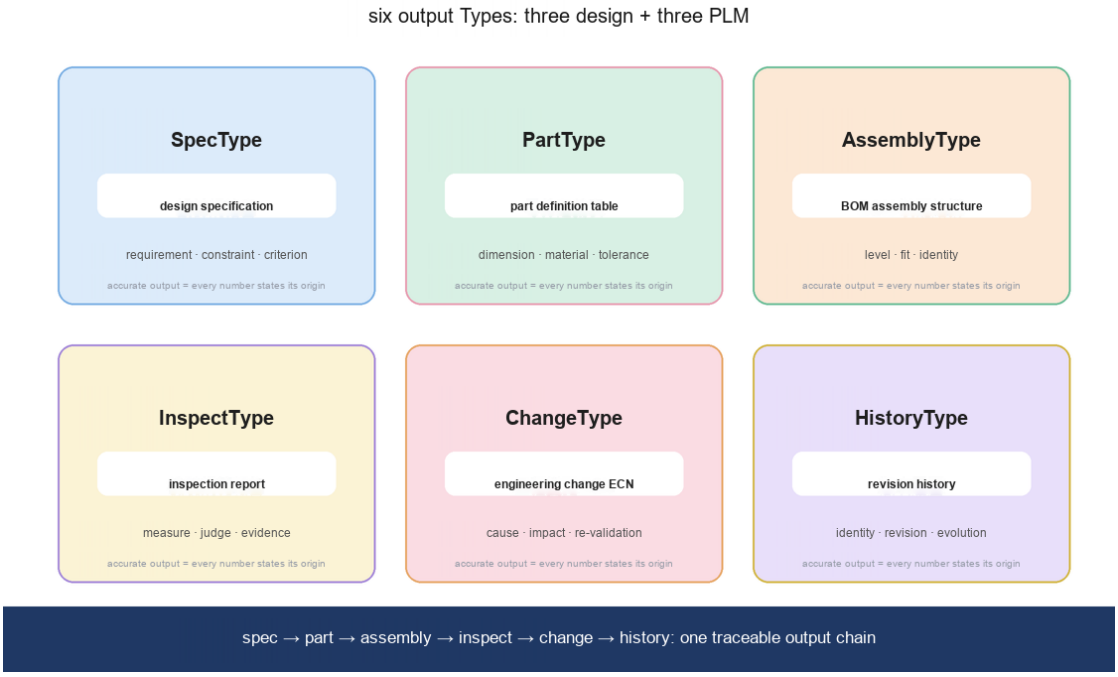


Figure 2-1 The six output Types: three design Types + three PLM Types

Table 2-1 Master table of the six output Types

Output Type	English	core question	key criterion
Spec type	SpecType	what is to be made	requirements quantified and acceptable
PartType	PartType	what this thing is	fields complete, values sourced
AssemblyType	AssemblyType	who goes inside whom	BOM criterion, correct fits
InspectType	InspectType	whether the measurement passes	criteria come from the SpecType
ChangeType	ChangeType	who is affected by the change	impact analysis complete, re-validation complete
HistoryType	HistoryType	past and present lives	id(C) never breaks, always traceable

Table 2-1 The six Types join into one output chain: one Type's output is the next Type's criterion.

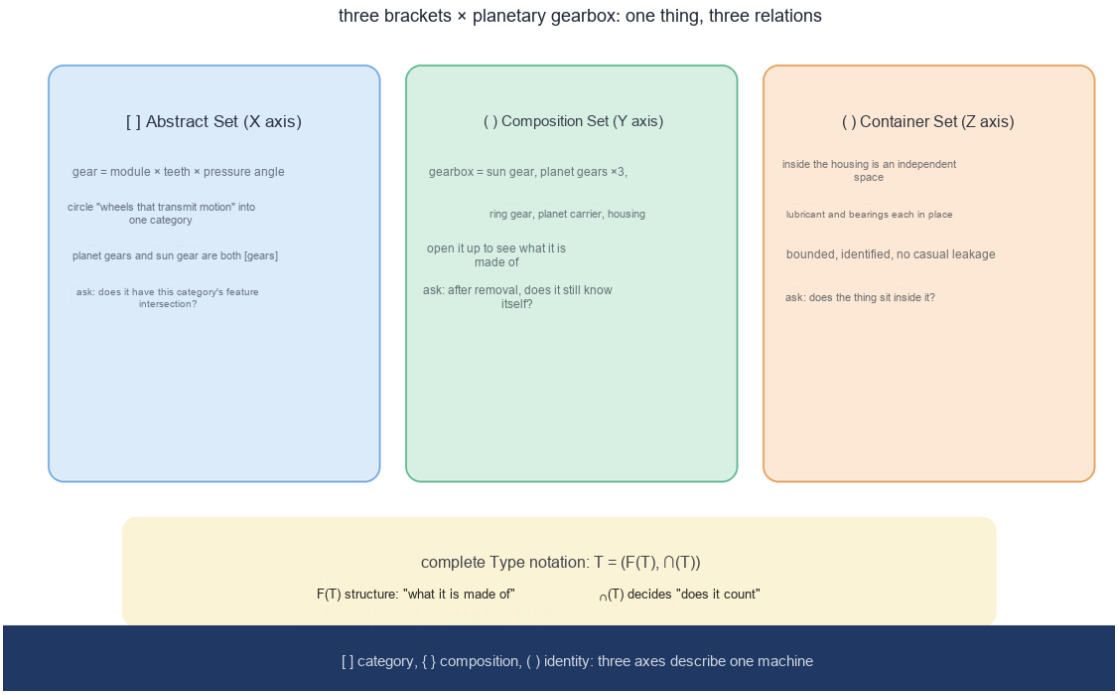


Figure 2-2 The three brackets × planetary gearbox: three relations of the same machine

2.1 Design Specification Type (SpecType)

The SpecType is the source of the output chain: the customer says "I want a small, high-torque gearbox", and the SpecType turns it into acceptable clauses. Its point is that every requirement is quantified into a criterion - a clause without a criterion is not a specification but a wish.

Table 2-2 Key fields of the design SpecType

Field	Description	planetary gearbox example
specification number	unique identity	SPEC-PG-001
requirement clause	the original requirement text	the customer requires a gear ratio of 5:1
quantified criterion ∩(T)	acceptable value ranges	i = 5 ± 2%;T2 ≥ 50 N·m
constraints	boundaries and limits	OD ≤ 120 mm; weight ≤ 3.5 kg
Acceptance method	how to judge fit	no-load run-in + rated torque test
traceability	which meeting / which quotation it came from	RFQ-2026-0815 clause 3

Table 2-2 The quantified criterion column is where ∩(T) lands: the SpecType carries its own acceptance standard.

2.2 Part Type (PartType)

The PartType is the leaf of the BOM: every part that can no longer be decomposed (for this case) becomes a Type. It records not only dimensions but material, heat treatment, tolerance and drawing number - miss one field and procurement, machining and inspection downstream each start guessing.

Table 2-3 Key fields of the PartType

Field	Description	sun gear example
part number	unique identity, the anchor point of id(C)	PT-SG-018
Name	conventional names	sun gear
material and heat treatment	determines strength and life	SCM415, carburised and quenched, HRC58-62
key dimensions	functional dimensions carry units	module 1.5; 18 teeth; pitch circle φ27 mm
tolerance	measurable accuracy requirements	tooth profile ISO 1328 grade 6; shaft diameter φ12 k6
surface treatment	corrosion resistance and appearance	black oxide treatment
figure number	two-way link with the drawing files	DWG-PG-018

Table 2-3 Every value must be tagged with its source when generated in S4 (see Chapter 5).

2.3 Assembly Type (AssemblyType)

The AssemblyType answers "who goes inside whom": BOM levels, quantities, fit relations, assembly sequence and tightening torque. This is exactly where the BOM criterion (after removal, does it still know itself?) and the three levels of remove earn their keep.

Table 2-4 Key fields of the AssemblyType

Field	Description	planetary train assembly example
assembly number	assembly identity	AS-PGS-01
BOM level	L0 machine / L1 assembly / L2 part	L1 (belongs to L0 planetary gearbox)
details and quantities	composition list	sun gear ×1, planet gears ×3, ring gear ×1, pins ×3
fit relation	shaft-bore fits and tolerance zones	planet gear bore φ14 H7 / pin φ14 k6
assembly sequence	reproducible steps	press the pin first, then fit the planet gears, then close the ring gear
tightening specification	torque and anti-loosening	M8 × 24 N·m + threadlocker

Table 2-4 The fit-relation column is the input to tolerance-chain analysis (see 3.3).

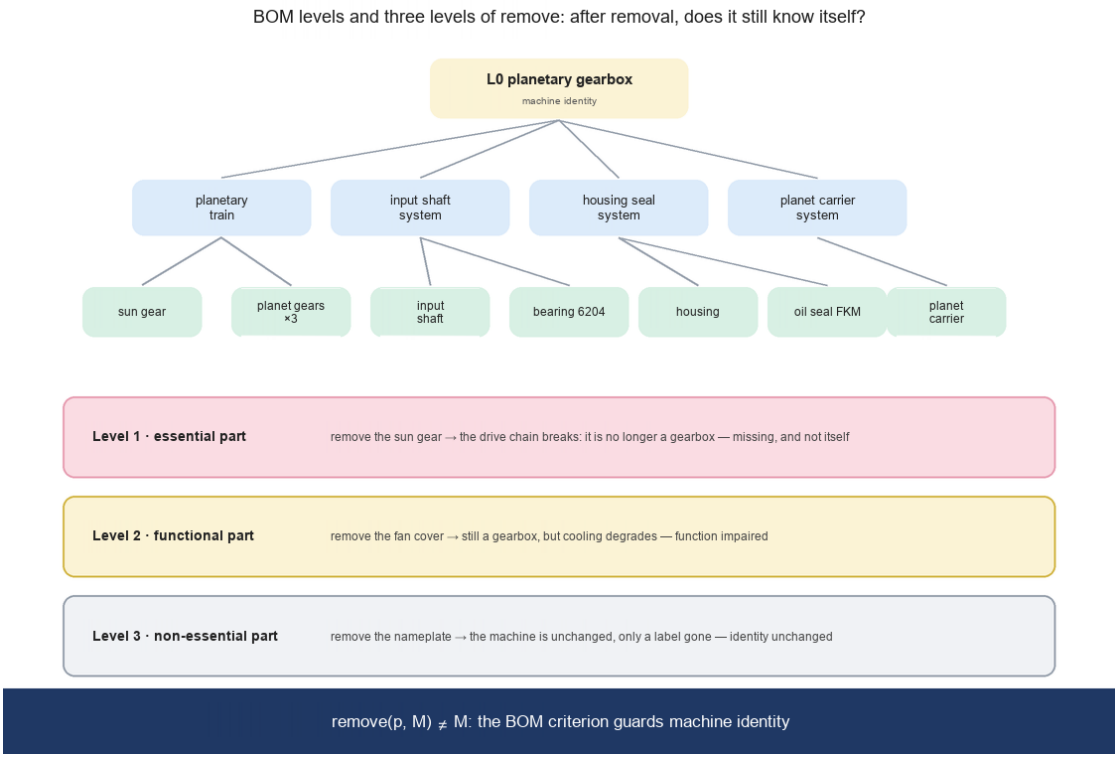


Figure 2-3 BOM levels and the three levels of remove: after removal, does it still know itself

Table 2-5 The three levels of remove × planetary gearbox

Grade	Criteria	example	how it is handled at change time
Level 1 - essential part	remove and it is no longer itself	sun gear, ring gear	must re-validate, must issue an ECN
Level 2 - functional part	remove and the function degrades	fan cover, oil cooler pipe	assess the degree of degradation before deciding the re-validation scope
Level 3 - non-essential part	remove and identity is unchanged	nameplate, appearance label	light change, does not trigger re-validation

Table 2-5 The three levels of remove are the computational basis of ChangeType impact analysis (see 2.5 and 5.2).

2.4 Inspection Type (InspectType)

The InspectType translates the SpecType's criteria into the language of measurement: inspection item, method, instrument, measured value, judgement. Every one of its criteria must trace back to the SpecType - the shop floor may not invent its own standard, and "roughly passing" is not allowed.

Table 2-6 Key fields of the InspectType

Field	Description	example
inspection item	what to measure	backlash
measurement method and instrument	how to measure	dial-indicator reversal method; resolution 0.01 mm
Criteria	which SpecType clause it comes from	SPEC-PG-001 §4:≤ 7 arcmin
measured value	with units and measurement conditions	0.09 mm (about 6.2 arcmin) @ 20°C
Determination	pass / fail / concession	Pass
record	report number and sign-off	INSP-PG-001-A / QA signature

Table 2-6 The judgement column allows only three values, shutting out the infinite space of "should be fine".

2.5 Change Type (ChangeType)

The ChangeType is the ECN (engineering change notice). It is not a notification but a small design exercise: state the cause of the change, before and after, the scope of impact and the re-validation items. The scope is computed from the AssemblyType dependency chain and the three levels of remove, not from the engineer's memory.

Table 2-7 Key fields of the ChangeType

Field	Description	ECN-042 example
change number	unique identity	ECN-042
cause	why the change	customer raises the upper limit of the operating temperature
change content	before-and-after comparison	oil seal material NBR → FKM
impact analysis	which Types are affected	PartType / AssemblyType / InspectType / SpecType
re-validation items	mandatory measurements and tests	150°C seal hold test, lip wear re-measurement
effective revision	when the new baseline starts	revision B (revision A archived)

Table 2-7 How the impact-analysis column is generated is shown in Figure 5-2; a full ECN example is in Table 5-7.

2.6 History Type (HistoryType)

The HistoryType is where the Time Type lands in PLM: every release, every ECN and every sign-off is collected onto one timeline. Its guardian is the identity anchor id(C) - revisions never break, and revisions A, B and C are the past and present lives of the same machine.

Table 2-8 Key fields of the HistoryType

Field	Description	example
revision	baseline code	Rev A → Rev B
release date	when it takes effect	2026-03-12
bound change	which ECNs this revision contains	Rev B = Rev A + ECN-042
release basis	under what conditions it is released	type tests all passed
Sign-off	who is responsible	joint sign-off by design, QA and manufacturing
archive	how the old revision is disposed of	Rev A drawing stamped void, filed for ten years

Table 2-8 The HistoryType keeps "why was it designed this way back then" permanently answerable.

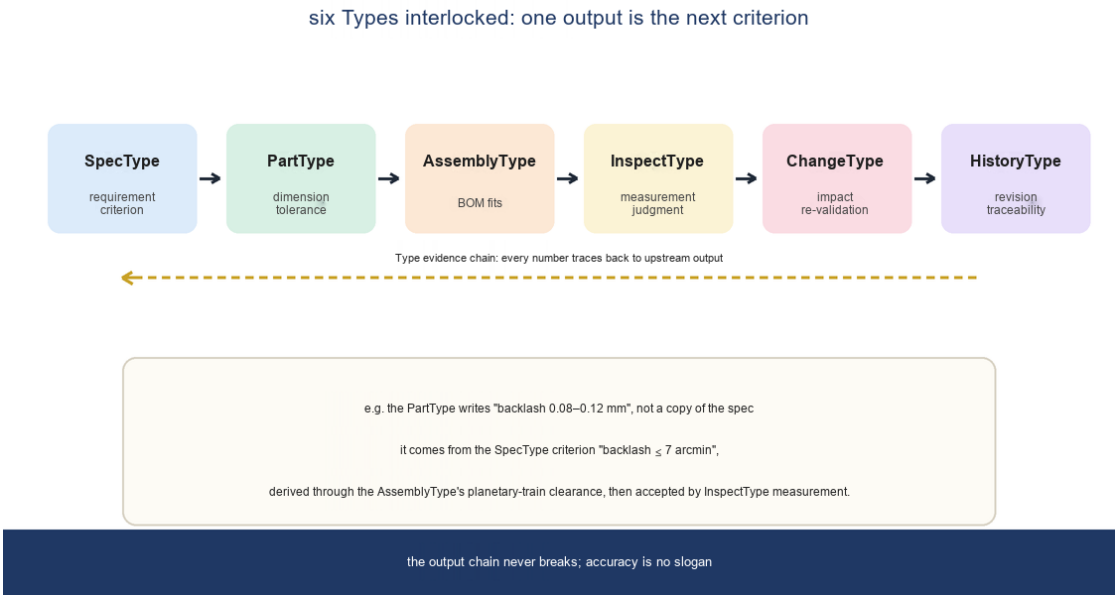
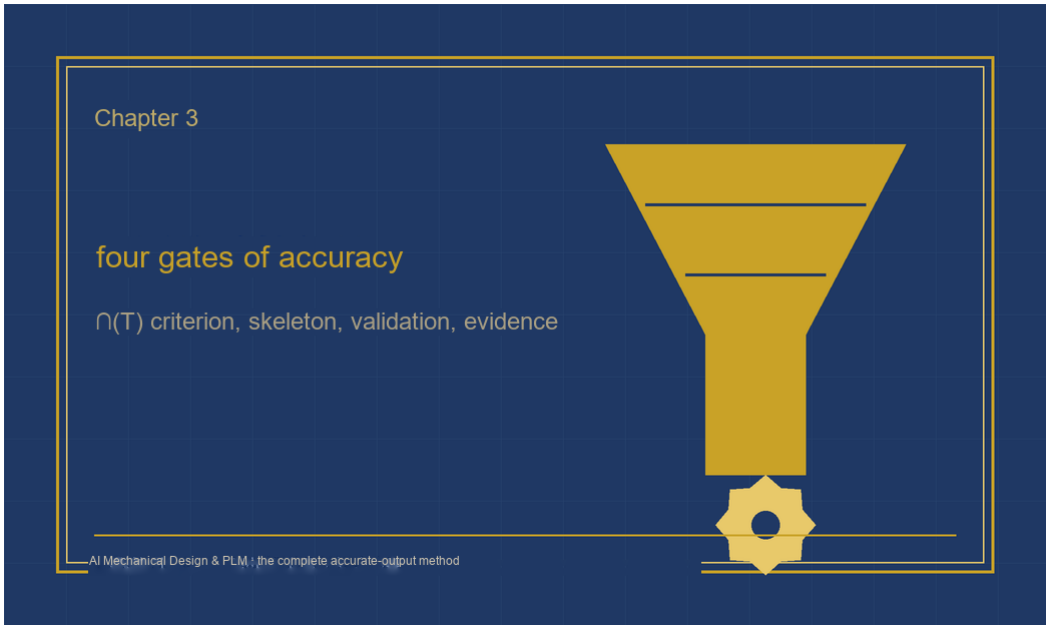


Figure 2-4 The six Types interlocked: one Type's output is the next Type's criterion, and the evidence chain traces backwards

Chapter focus

Six output Types = spec, part, assembly, inspection, change, history: the three design Types (specification, part, assembly) plus the three PLM Types (inspection, change, history).
The quantified criterion of the SpecType is ∩(T): a clause without a criterion is not a specification.
The three levels of remove turn "how big is the impact" into a computable question.
The HistoryType keeps identity continuous through id(C): revisions never break.

Chapter 3 The Four-Layer Accuracy Guarantee



The previous chapter set up the six output Types; this chapter answers "on what grounds do we call it accurate". The answer is four gates: the ∩(T) intersection criterion guards "does it count", the deductive skeleton guards "was it derived", engineering validation guards "was it tested", and the Type evidence chain guards "can it be found". The four run from entrance to exit; if any gate fails, the output is downgraded to "for reference only".

Table 3-1 Master table of the four gates of accuracy

Layer	what it protects	Tool	symptoms of failure
First layer	eligibility: does it count as this Type	∩(T)	accept everything and the Type Library becomes a dump
Second layer	reasoning: how the conclusion was reached	Deductive skeleton	the number is right but its provenance is unknown
Third layer	reality: can the numbers hold up	engineering validation	passes on paper, fails in the field
Fourth layer	memory: can it be found afterwards	Type evidence chain	the same error repeats and nobody owns it

Table 3-1 The four gates stack: all must pass for accuracy; missing one leaves it incomplete.

3.1 Gate One: the ∩(T) Intersection Criterion

Every Type carries an intersection criterion ∩(T): the intersection of a set of necessary conditions, and only output satisfying all of them enters the Type. In mechanical design ∩(T) becomes the four-gate funnel - requirement fit, catalogue availability, standard compliance, criterion acceptance - every gate filtering, every gate keeping evidence.

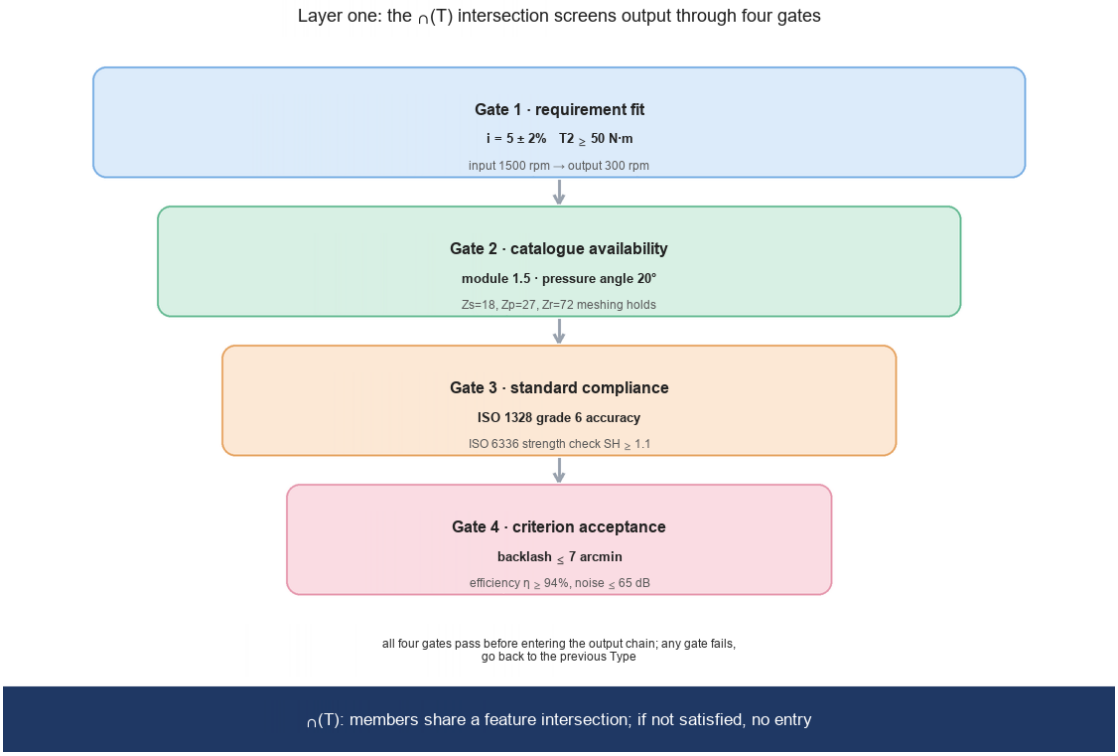


Figure 3-1 The four-gate ∩(T) funnel: every gate filters, every gate keeps evidence

Table 3-2 Example of the four-gate ∩(T) criteria (planetary gearbox)

Gate	criterion content	Action when not satisfied
requirement fit	i = 5 ± 2%;T2 ≥ 50 N·m;n1 ≤ 3000 rpm	return to S1 and rewrite the SpecType
catalogue availability	module and tooth counts satisfy the meshing conditions; (Zs+Zr)/number of planet gears is an integer	return to S2 and reselect the tooth parameters
standard compliance	ISO 1328 grade 6; ISO 6336 check SH ≥ 1.1	return to S3 and increase the module or change the material
Criteria acceptance	backlash ≤ 7 arcmin; η ≥ 94%; noise ≤ 65 dB	return to S5 and open a non-conformance disposition

Table 3-2 The order of the four gates cannot be swapped: requirements first, then selection, then standards, acceptance last.

The value of ∩(T) is that it holds the line. An output chain without criteria is just an elegant guess, however fluently the AI writes it; with criteria the guess is stopped at the first gate - and that is the first door to "accurate output".

3.2 Gate Two: the Deductive Skeleton

The second gate governs the reasoning process: every conclusion must be derived along the skeleton, every step being "known → formula → substitution → result". The opposite of the deductive skeleton is copying the answer: the answer happens to be right but nobody can check the process - change the operating condition next time and it is all wrong.

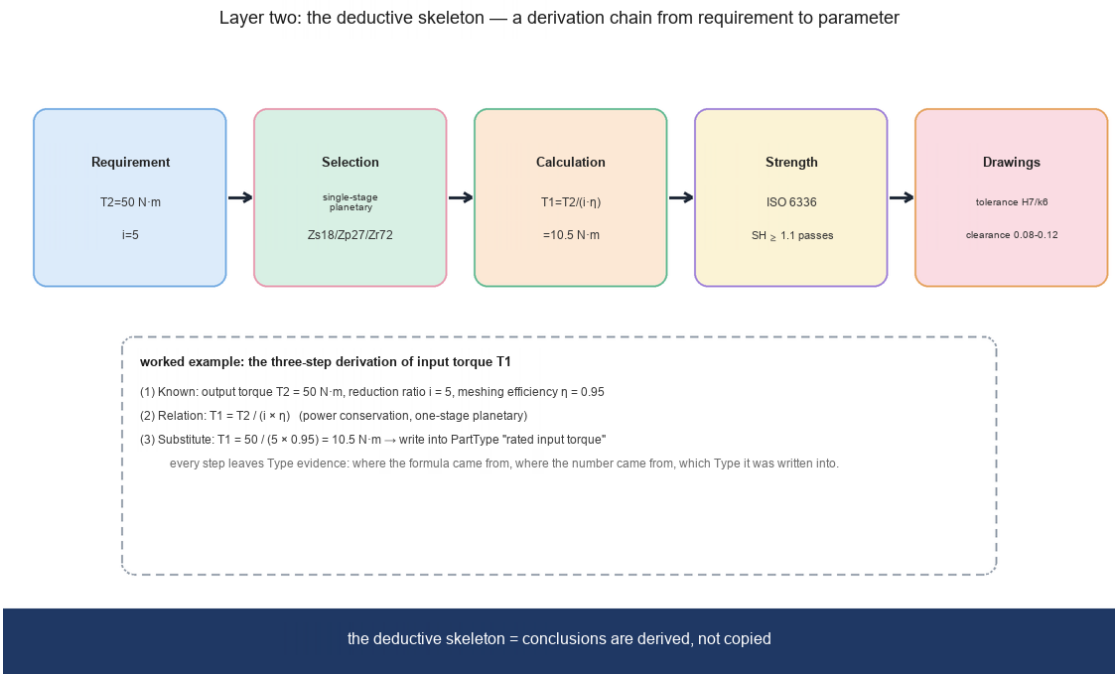


Figure 3-2 The deductive skeleton: the derivation chain from requirement to parameter

The input torque of the planetary gearbox is a typical derivation: given output torque T2 = 50 N·m, ratio i = 5, meshing efficiency η = 0.95; conservation of power gives T1 = T2 / (i × η); substituting gives T1 = 50 / (5 × 0.95) = 10.5 N·m. This result is then written into the "rated torque of input shaft" field of the PartType and its formula source is hung on the Type evidence chain. The meshing condition of the gears works the same way: for a single-stage planetary drive i = 1 + Zr/Zs, so Zr/Zs = 4, take Zs = 18 and Zr = 72; the concentricity condition Zr = Zs + 2Zp then gives Zp = 27; the assembly condition (Zs + Zr) / 3 = 30 is an integer, so three planet gears equally spaced holds.

Table 3-3 Deductive skeleton derivation table (excerpt)

Steps	Known	Relation	Result
(1)	T2 = 50 N·m;i = 5;η = 0.95	T1 = T2 / (i x η)	T1 = 10.5 N·m
(2)	i = 1 + Zr / Zs	Zr = 4 Zs	Zs = 18,Zr = 72
(3)	concentricity condition Zr = Zs + 2 Zp	Zp = (Zr - Zs) / 2	Zp = 27
(4)	assembly condition (Zs + Zr) / n is an integer	n = 3 equally spaced	90 / 3 = 30 holds
(5)	P = T2 x n2 / 9550	n2 = n1 / i = 300 rpm	P = 1.57 kW

Table 3-3 Every result carries units and every formula has a source - that is the point of the skeleton.

3.3 Gate Three: Engineering Validation

Correct derivation is not enough: the numbers must survive reality. The third gate is engineering validation - tolerance stack-up, strength checks and performance tests, nailing the paper numbers down with measurements from the physical world. For AI output this gate is the watershed of "dare we stamp it".

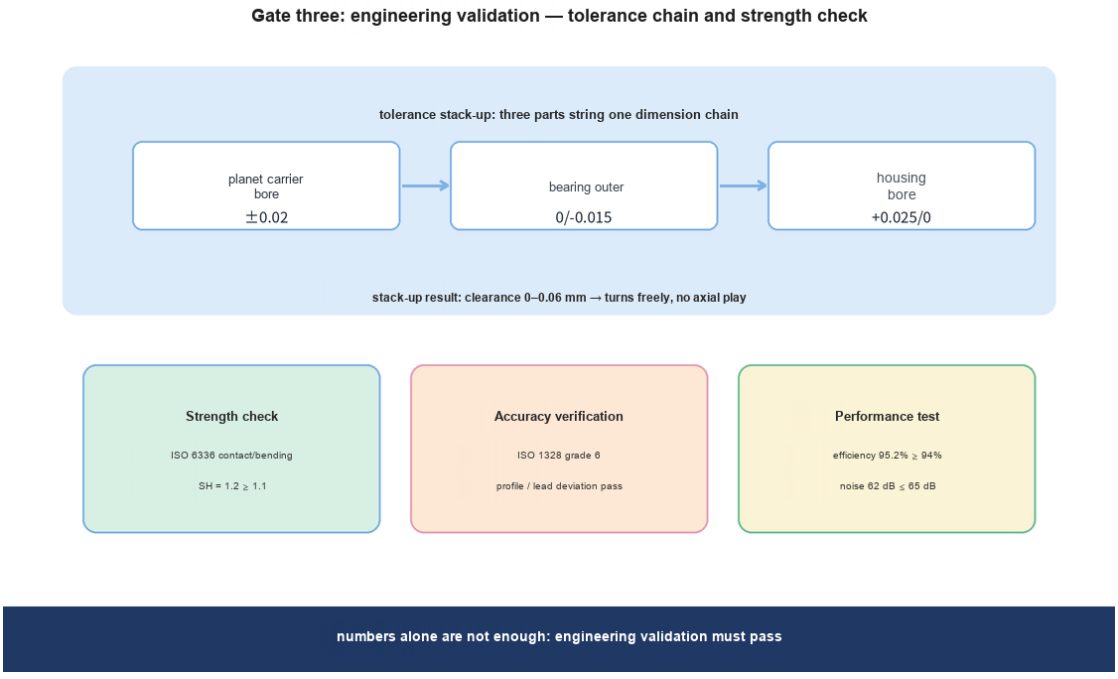


Figure 3-3 Engineering validation: tolerance stack-up plus the three-way checks of strength and performance

Table 3-4 Tolerance-chain analysis table (radial location of the planetary train)

composition ring	tolerance	worst-case stack-up	Result
housing bore	+0.025 / 0	max 120.025	—
bearing outer diameter	0 / -0.015	min 59.985	—
planet carrier bore	±0.02	max 120.02	—
◆Gap stack-up	—	0 ~ 0.06 mm	**◆Pass**

Table 3-4 The worst-case method looks at the worst situation first; tighten later in mass production with statistical methods (Cp/Cpk).

Each of the three checks owns one question: strength checking (ISO 6336 contact and bending, safety factors SH ≥ 1.1 and SF ≥ 1.25) asks "will it break"; accuracy verification (ISO 1328 profile and lead deviations) asks "will it run smoothly"; performance testing (efficiency, noise, temperature rise) asks "is it good to use". If any one fails, the output goes back to the corresponding Type to be redone - not patched in the report.

3.4 Gate Four: the Type Evidence Chain

The last gate handles memory and traceability: the Type evidence chain writes "one source per number, one ledger per Type" into the output. Behind every value on a PartType hangs its birth certificate - which specification clause it came from, which formula it travelled, which measurement it passed. A number without evidence may not be booked.

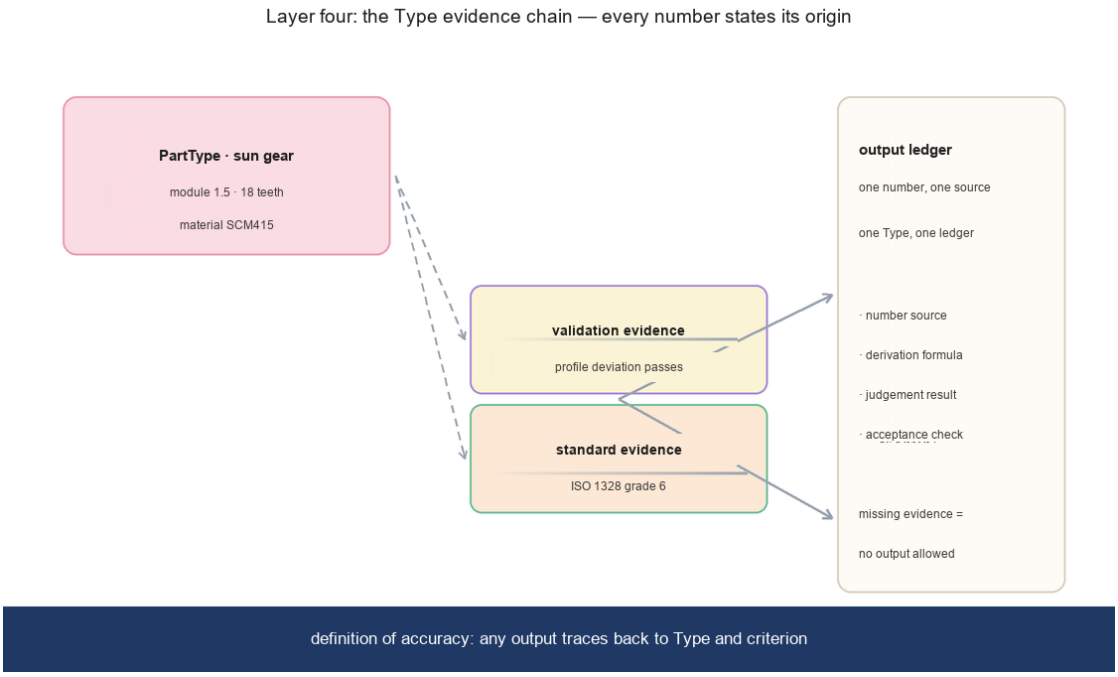


Figure 3-4 The Type evidence chain: every number can state its origin

Table 3-5 Type evidence chain format (PartType excerpt)

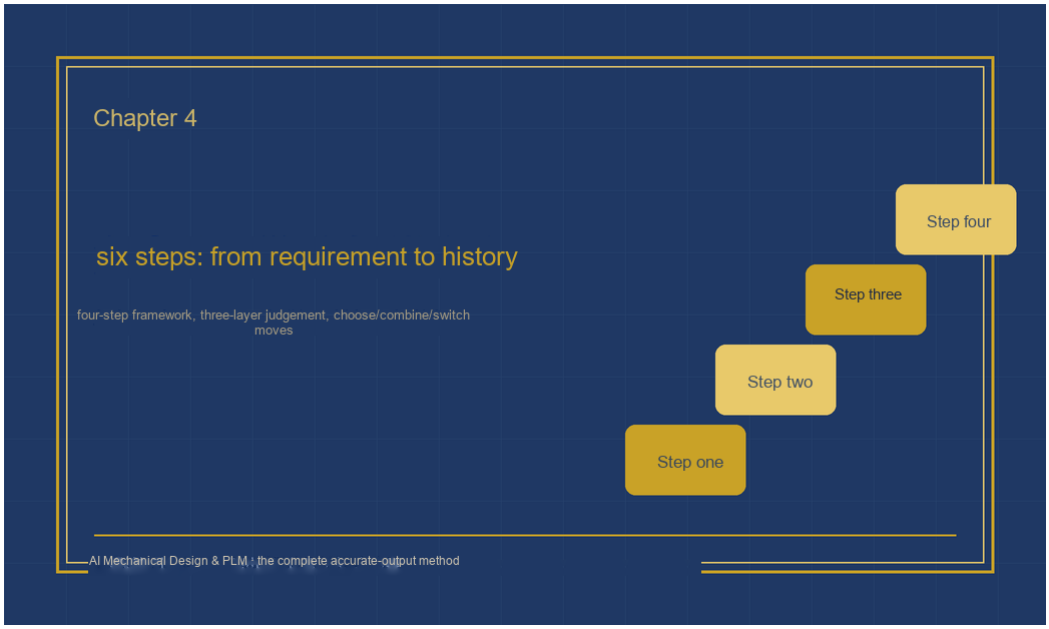
value	source Type	derivation / basis	validation record
18 teeth	SpecType SPEC-PG-001 §2	back-calculate with i = 1 + Zr/Zs	—
material SCM415	SpecType SPEC-PG-001 §6	contact-stress material selection table	material certificate CM-0415
shaft diameter φ12 k6	AssemblyType AS-PGS-01	shaft-bore fit H7/k6	INSP-PG-001-A §2
backlash 0.08-0.12 mm	SpecType §4 backlash criterion	backlash conversion	INSP-PG-001-A §5

Table 3-5 Space × time cross acceptance: in space it traces back to the Type, in time to the revision and the measurement records.

Chapter focus

The four gates run from entrance to exit: criteria → derivation → validation → traceability.
The four-gate ∩(T) funnel: requirement fit, catalogue availability, standard compliance, criterion acceptance.
The deductive skeleton refuses copied answers: known → formula → substitution → result, units at every step.
The Type evidence chain, one source per number: a number without evidence may not be booked.

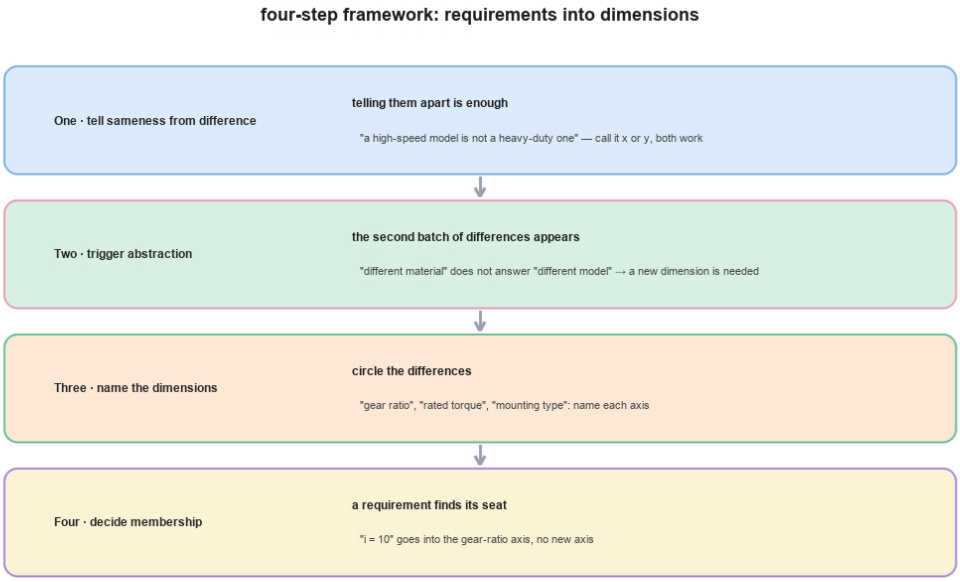
Chapter 4 The Six Steps: From Requirement to History



The first two chapters gave the output Types and the insurance on accuracy; this chapter gives the method of working - how to take a case from requirement to history. The method has three layers: the four-step framework turns requirements into dimensions, the six steps walk the case one full round, and the three-layer judgement plus move combinations find a way when you are stuck.

4.1 The Four-Step Framework: Turning Requirements into Dimensions

When a pile of requirements lands, do not start drawing. The four-step framework - one, tell sameness from difference; two, trigger abstraction; three, name the dimension; four, decide membership - sorts the mess into dimensions, so that every requirement has somewhere to hang.



framework = what is left after it runs: a reusable skeleton

Figure 4-1 The four-step framework: from messy requirements to ordered dimensions

Table 4-1 The four-step framework × requirement analysis

Steps	trigger timing	planetary gearbox example
One: distinguish similarity and difference	a cluster of requirements arrives	"A high-speed model is not a heavy-duty model" - first tell them apart
Two: trigger abstraction	the second batch of differences does not match	"Different mounting flange" does not answer "different gear ratio"
Parameter, naming the dimension	differences must be understood separately	set up the axes of ratio, rated torque, mounting type and so on
Four: judging membership	a new requirement arrives	"Want i = 10" hangs on the transmission-ratio axis; no new axis is opened

Table 4-1 The framework is what is left after the framework has run: a skeleton that can be applied again and again.

4.2 The Six Steps End to End

With the dimensions set, the case goes one round of the six steps: identify the situation, build Types by axis, select and expand the pattern, generate the output, criterion acceptance, archiving and evolution. Each step has clear entry conditions, actions and outputs - after six steps the six Types are complete and the case can be closed.

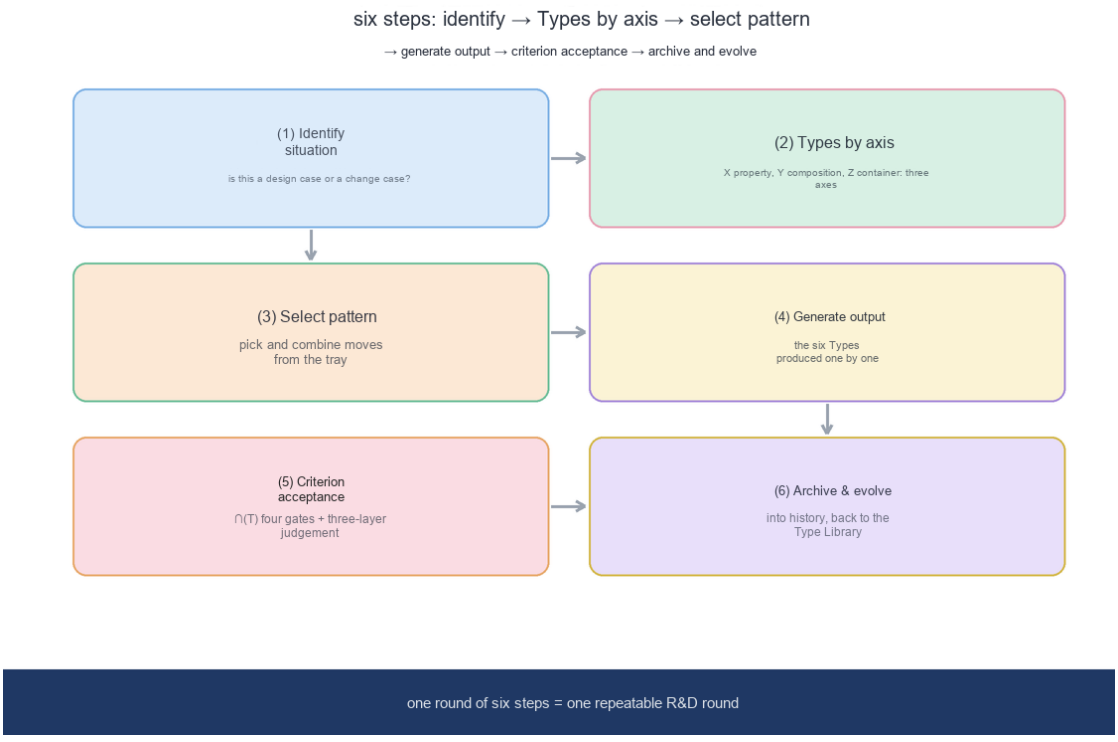


Figure 4-2 The six steps: one repeatable R&D round

Table 4-2 The six steps × actions and outputs

Steps	key actions	Output Type	completion criterion
(1) Identify the situation	routing design cases / change cases / quotation cases	—	situation identified, scope bounded
(2) Build Types by axis	X property, Y composition, Z container: the three axes	SpecType draft	dimensions named
(3) Select and expand the pattern	choose and combine moves from the move tray	—	moves match the situation

Steps	key actions	Output Type	completion criterion
(4) Generate the output	S1-S7 pipeline operation	PartType - AssemblyType	fields complete, values sourced
(5) Criterion acceptance	the four ∩(T) gates + engineering validation	InspectType	all judged pass
(6) Archiving and evolution	enter the history, feed back to the Type Library	ChangeType - HistoryType	release the revision, accumulate experience

Table 4-2 The mapping between the six steps and S1-S7 is detailed in Chapter 5; the move tray for (3) is in Table A-2.

4.3 Three-Layer Judgement

Day-to-day work constantly needs the judgement "does this count": is this equipment a container? which level is this part? is this new product a gearbox? The three-layer judgement of Type logic answers in order: first the container identity (( )), then the composition structure ({ }), and finally the abstract category ([ ]). Get the order right and half the disputes disappear.

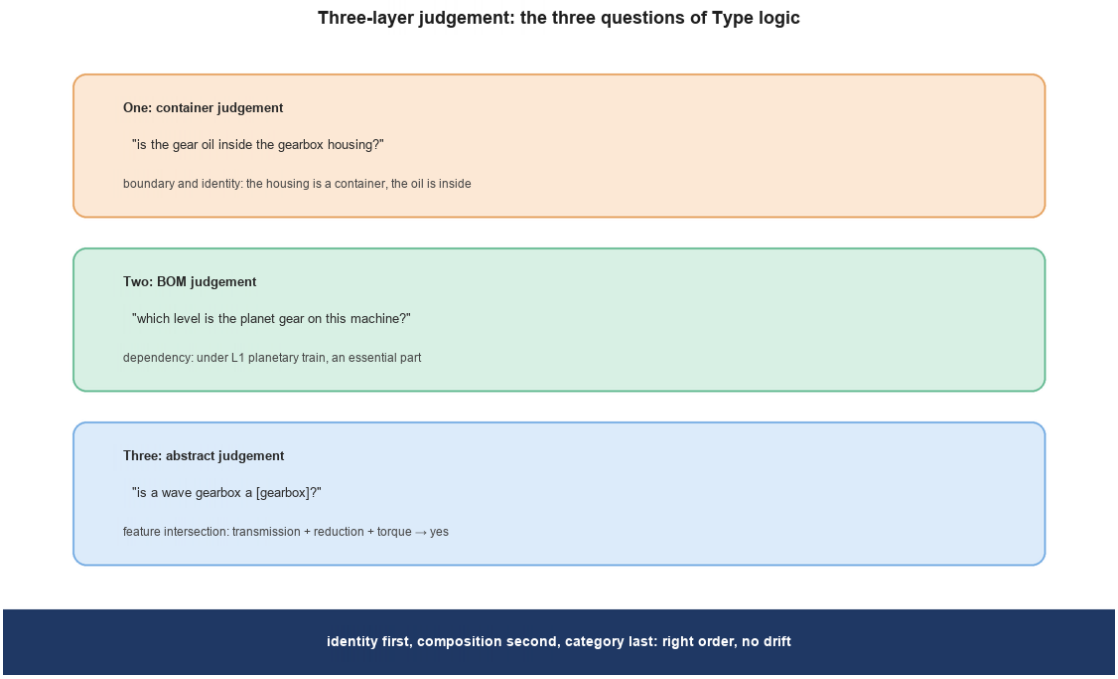


Figure 4-3 The three questions of Type logic: identity first, composition second, category last

4.4 Choosing, Combining and Switching Moves

The move tray of the Time Type must be used flexibly: picking a single move by situation is choosing; stringing several into a routine is combining; switching to another path when stuck is switching. A heavy-duty design case can start by choosing "design search and extraction", then combine "analysis → design → reasoning" to push forward; if forward design will not move, switch to "reverse thinking" or "first-principles thinking" and re-decompose.

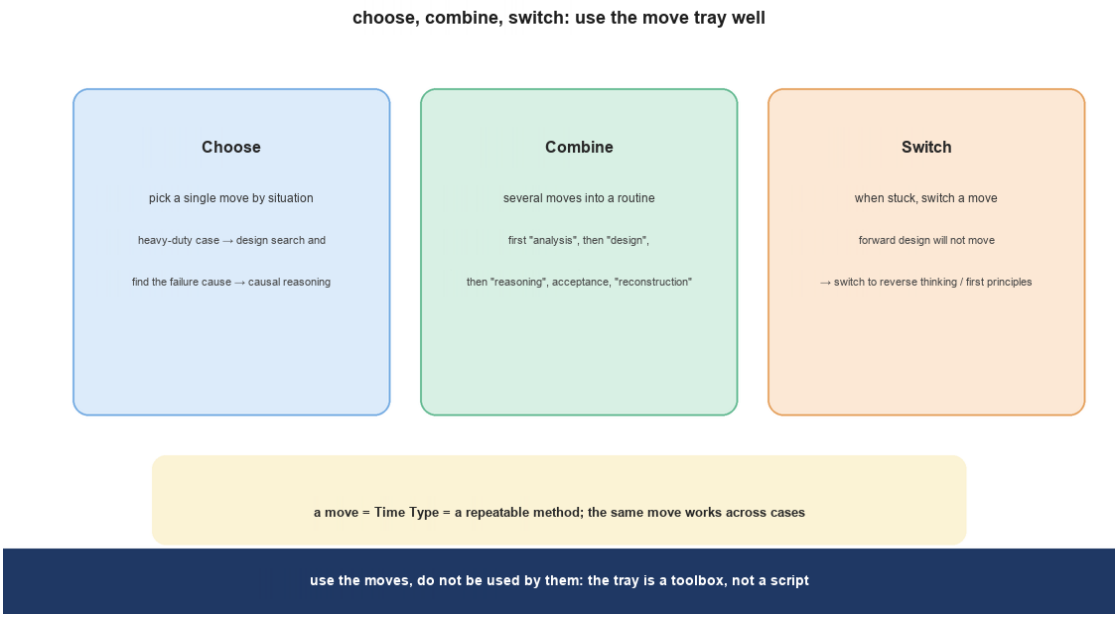


Figure 4-4 Choosing, combining and switching moves: using the move tray flexibly

Table 4-3 Examples of move operations

Operation	Situation	combine
Selecting moves	the customer wants "more torque in the same volume"	design search and extraction (find the existing limit)
Assembling moves	development of an entirely new transmission architecture	analysis → design (creation and innovation) → reasoning (strength causality) → refactoring
Swapping moves	parameter optimisation stuck at a local optimum	forward thinking → switch to the level-thinking method / first principles to re-establish the dimensions

Table 4-3 A move is a method, not a script: the input is a situation, the output is a document, and reuse is its duty.

Chapter focus

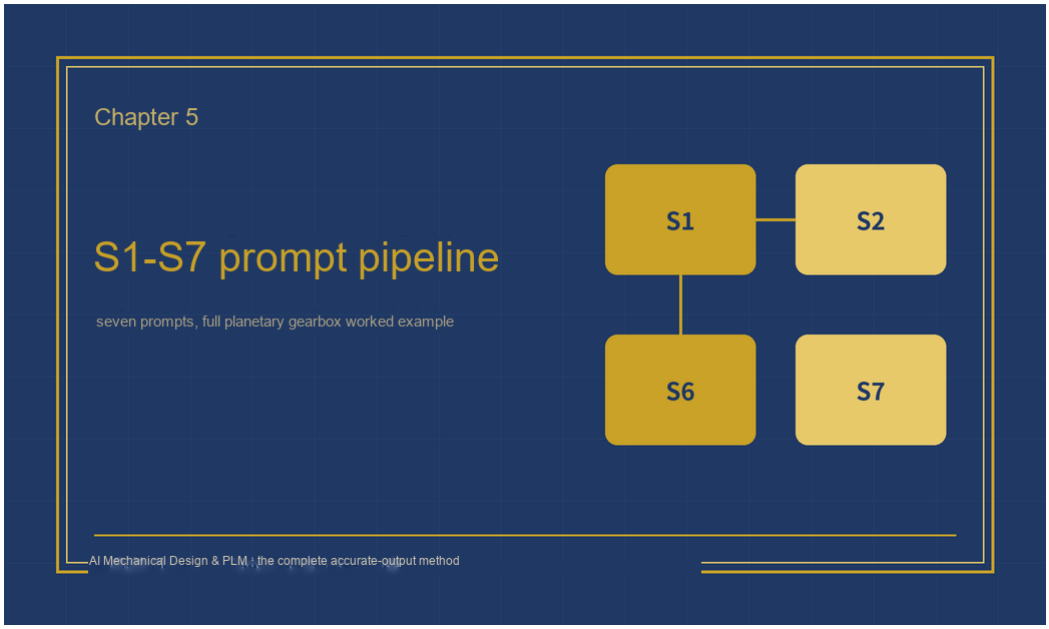
The four-step framework turns requirements into dimensions: tell sameness from difference → trigger abstraction → name the dimension → decide membership.

One round of the six steps: identify the situation → build Types by axis → select and expand the pattern → generate the output → criterion acceptance → archiving and evolution.

The three-layer judgement has a fixed order: container → composition → abstract.

Moves must be used flexibly: choose, combine, switch.

Chapter 5 The S1-S7 Prompt Pipeline and a Full Worked Example



This chapter is the operating core of the book. The first half is the full text of the seven-stage S1-S7 prompt pipeline - each stage states role, input, action, output Type and acceptance criterion; the second half walks the pipeline from the beginning with the planetary gearbox, showing what the complete output of the SpecType, PartType, AssemblyType, InspectType, ChangeType and HistoryType actually looks like.

5.1 The S1-S7 Seven-Stage Prompt Pipeline

The pipeline's design principle: each stage does one thing only, produces one Type only, and answers to one set of criteria only. Stages hand over through the output Type, so any stage can be re-run without contaminating the others.

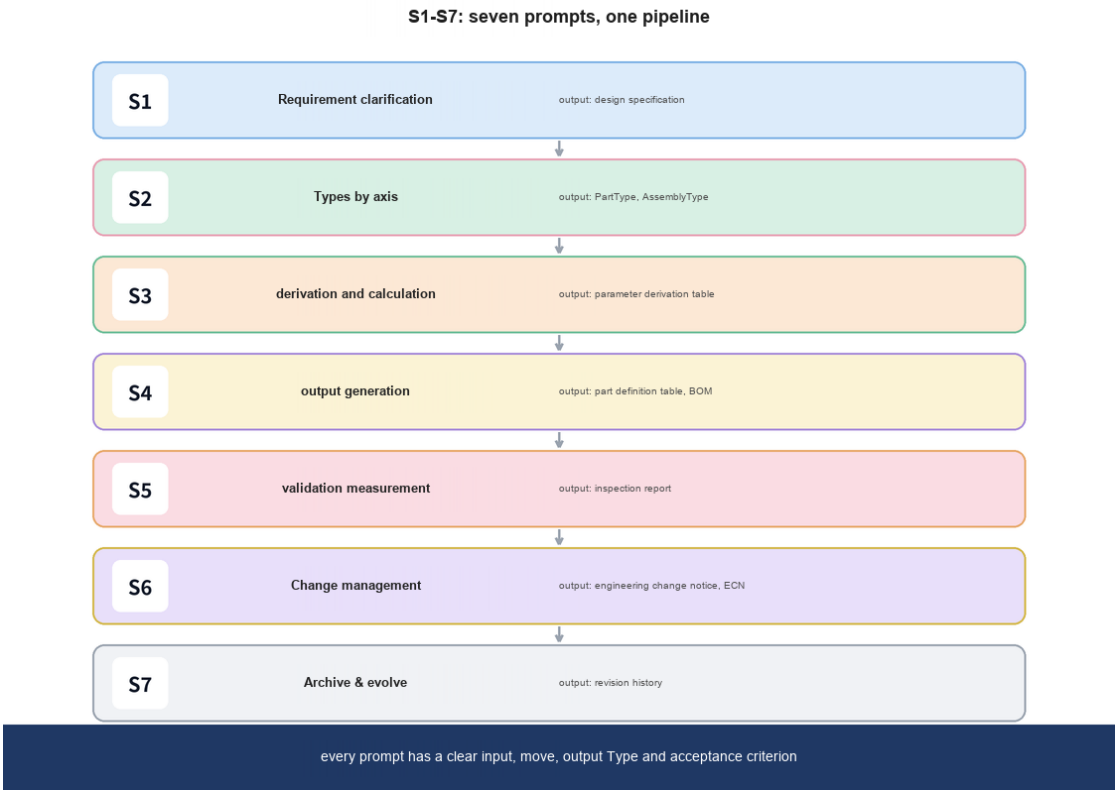


Figure 5-1 The S1-S7 pipeline: every stage has an input, a move, an output Type and an acceptance criterion

Table 5-1 Master table of S1-S7

Stage	Task	main moves	Output Type	Acceptance criteria
S1	Clarifying the requirement	Analysis, problem solving	Spec type	every requirement is quantified
S2	Build Types by axis	Four framework steps	PartType - AssemblyType	all three axes complete, remove grading done
S3	derivation and calculation	reasoning (causal)	parameter derivation table	every step has units and a formula
S4	output generation	Function generation and application	part definition table, BOM	every value is tagged with its source
S5	validation and measurement	reasoning (simulation)	InspectType	all criteria come from the SpecType
S6	change management	Refactoring and recombination	ChangeType	impact analysis covers the whole dependency chain
S7	Archive and evolve	Continuous learning, self-evolution	HistoryType	release the revision, feed back to the Type Library

Table 5-1 The move column maps to where the fourteen moves land in Table A-2 of Appendix A.

S1 - Requirement Clarification

S1 prompt (requirement clarification → SpecType)

You are a mechanical design engineer. Organise the following customer requirements into a [SpecType] design specification.
Requirements: (1) list the original text of each requirement clause one by one; (2) quantify each into an acceptable criterion ∩(T) (value + range);
(3) Annotate the constraints (dimensions, weight, environment, regulations); (4) state the acceptance method.
Never fabricate values the customer did not mention; mark gaps [To Be Confirmed] and list the clarification questions.
Output fields: specification number | requirement clause | quantified criterion | constraints | acceptance method | traceability.
Customer requirement: {customer description}

S2 - Building Types by Axis

S2 prompt (build Types by axis → PartType and AssemblyType)

Using SpecType SPEC-xxx, build the product Type along the three axes.
X axis [ ]: list the key properties and their values (name | value | unit | source clause).
Y axis { }: expand the BOM down to part level (level L0-L2 | number | name | quantity).
Z axis ( ): mark the container boundaries (housing / module / installation space, with id(C) and boundary description for each container).
And grade every component with the three levels of remove: essential / functional / non-essential part.
Output: AssemblyType skeleton + PartType list (fields may be left empty for S4 to fill in values).

S3 - Derivation and Calculation

S3 prompt (derivation and calculation → parameter derivation table)

Derive the main parameters step by step along the deductive skeleton. Each step outputs four columns: known | formula | substitution | result (with units).
where a standard is cited, give its number and clause (e.g. ISO 6336-1 contact strength).
Never give a conclusion without the derivation; if a parameter has several paths, take one as the main line and list the rest as alternatives.
Finally output the parameter derivation table, marking which PartType field each result will be written into.

S4 - Output Generation

S4 prompt (output generation → part definition table and BOM)

Generate the part definition table and the BOM from the Types and the derivation results.
Part fields: part number | name | material | heat treatment | key dimensions and tolerances | surface treatment | drawing number.
Tag the source after every value with [Source: ...] (specification clause number or derivation step number).
BOM fields: level | number | name | quantity | fit relation | remove level.
A value with no source may not be booked; anything of unclear origin is marked [Evidence Pending].

S5 - Validation and Measurement

S5 prompt (validation and measurement → InspectType)

Write the inspection specification and report. Output item by item: inspection item | measurement method | instrument | criterion (citing the SpecType clause number) |
measured value (simulated values may be marked [Simulated] first) | judgement (pass / fail / concession).
Every non-conforming item must come with a disposition recommendation and an automatically drafted change request (handed to S6).
Never invent criteria on the spot; vague judgements such as "should be fine" are banned.

S6 - Change Management

S6 prompt (change management → ChangeType ECN)

Write the engineering change notice ECN as a ChangeType.
Content: change number | cause | before-and-after comparison | scope of impact | mandatory re-validation items | effective revision.
The scope of impact must be computed from the three levels of remove and the AssemblyType dependency chain, and listed Type by Type (PartType / AssemblyType / InspectType / SpecType).
Changes to level-1 essential parts are automatically queued for mandatory re-validation; level-3 non-essential parts take the light-change route.

S7 - Archiving and Evolution

S7 prompt (archiving and evolution → HistoryType)

Archive this case as a HistoryType: revision | date | release basis | bound ECN list | sign-off status | archiving disposition.

and feed three things back to the Type Library: (1) the list of reusable PartTypes and AssemblyTypes; (2) the moves worth fixing and the prompt revisions;
(3) Misuses in this case and how they were corrected (see the five misuses in Chapter 6).
Finally output the suggested starting point for the next case (which Type in the Type Library to start from).

5.2 Full Worked Example: Planetary Gearbox

Case background: customer RFQ-2026-0815 asks for a single-stage planetary gearbox - input 1500 rpm, ratio 5:1, rated output torque 50 N·m, continuous duty, outer diameter ≤ 120 mm, operating environment -20 ~ 120°C. The output is demonstrated stage by stage below along S1-S7 (key fields excerpted; the full documents come out of the pipeline, and what is shown here is what really matters for acceptance).

Table 5-2 S1 output - design specification (SpecType excerpt)

specification number	requirement clause	quantified criterion ∩(T)	Acceptance method
SPEC-PG-001 §1	gear ratio 5:1	i = 5 ± 2%	no-load speed ratio measurement
SPEC-PG-001 §2	rated output torque	T2 ≥ 50 N·m (continuous)	magnetic-particle brake load test
SPEC-PG-001 §3	outer diameter and weight	OD ≤ 120 mm; ≤ 3.5 kg	instruments and scales
SPEC-PG-001 §4	transmission accuracy	backlash ≤ 7 arcmin	dial-indicator reversal method
SPEC-PG-001 §5	efficiency and noise	η ≥ 94%;≤ 65 dB(A)	back-to-back efficiency test
SPEC-PG-001 §6	operating environment	-20 ~ 120°C continuous operation	run-in in a thermal chamber

Table 5-2 Every clause is quantified and acceptable - wishes that do not qualify are marked [To Be Confirmed] right in S1.

Table 5-3 S3 output - main parameter derivation (excerpt)

Known	formula	Substitute	Result
T2 = 50 N·m;i = 5;η = 0.95	T1 = T2 / (i x η)	50 / 4.75	10.5 N·m
n1 = 1500 rpm;i = 5	n2 = n1 / i	1500 / 5	300 rpm
i = 1 + Zr / Zs	Zr = 4 Zs	Zs = 18	Zr = 72
Zr = Zs + 2 Zp	Zp = (72 - 18) / 2	54 / 2	Zp = 27
P = T2 x n2 / 9550	50 x 300 / 9550	—	P = 1.57 kW

Table 5-3 Verification of meshing and assembly conditions is in Table 3-3; every result is written into the corresponding PartType.

Table 5-4 S2/S4 output - BOM (AssemblyType excerpt)

level	No.	Name	quantity	remove level
L0	AS-PG-000	planetary gearbox	1	—
L1	AS-PGS-01	planetary train assembly	1	essential
L1	AS-IN-01	input shaft assembly	1	essential
L1	AS-HS-01	housing seal assembly	1	essential
L2	PT-SG-018	sun gear	1	essential
L2	PT-PL-027	planet gear	3	essential
L2	PT-RG-072	ring gear	1	essential
L2	PT-SF-006	oil seal FKM	2	essential
L2	PT-NP-001	nameplate	1	non-essential

Table 5-4 The fan cover (functional part) and others are omitted; the remove level is exactly the weight used in S6 impact analysis.

Table 5-5 S4 output - part definition table (PartType PT-SG-018 sun gear)

Field	Value	[Source]
material	SCM415, carburised and quenched, HRC58-62	SPEC §6 + material selection table S3-(5)
module / tooth count	1.5 / 18	SPEC §1;S3-(2)
pitch circle	φ27 mm(d = mZ)	S3 derivation step (2)
shaft diameter tolerance	φ12 k6	AssemblyType AS-PGS-01 fit table
gear accuracy	ISO 1328 grade 6	SPEC §4
surface treatment	black oxide treatment	SPEC §6 environmental clause
figure number	DWG-PG-018	S4 drawing list

Table 5-5 The third column is the Type evidence chain: a number missing [Source] is stopped at S4.

Table 5-6 S5 output - inspection report (InspectType INSP-PG-001-A excerpt)

inspection item	instrument / method	criterion (source)	Measured	Determination
gear ratio	tachometer back-to-back test	i = 5 ± 2%(SPEC §1)	5.01	Pass
backlash	dial-indicator reversal method	≤ 7 arcmin(SPEC §4)	6.2 arcmin	Pass
transmission efficiency	the back-to-back loading method	≥ 94%(SPEC §5)	95.2%	Pass
noise	sound level meter at 1 m	≤ 65 dB(A)(SPEC §5)	62 dB(A)	Pass
high-temperature sealing	150°C hold for 4 h	no leakage (ECN-042)	no leakage	Pass

Table 5-6 The criterion column cites SpecType clauses throughout - the measurement floor does not invent standards.

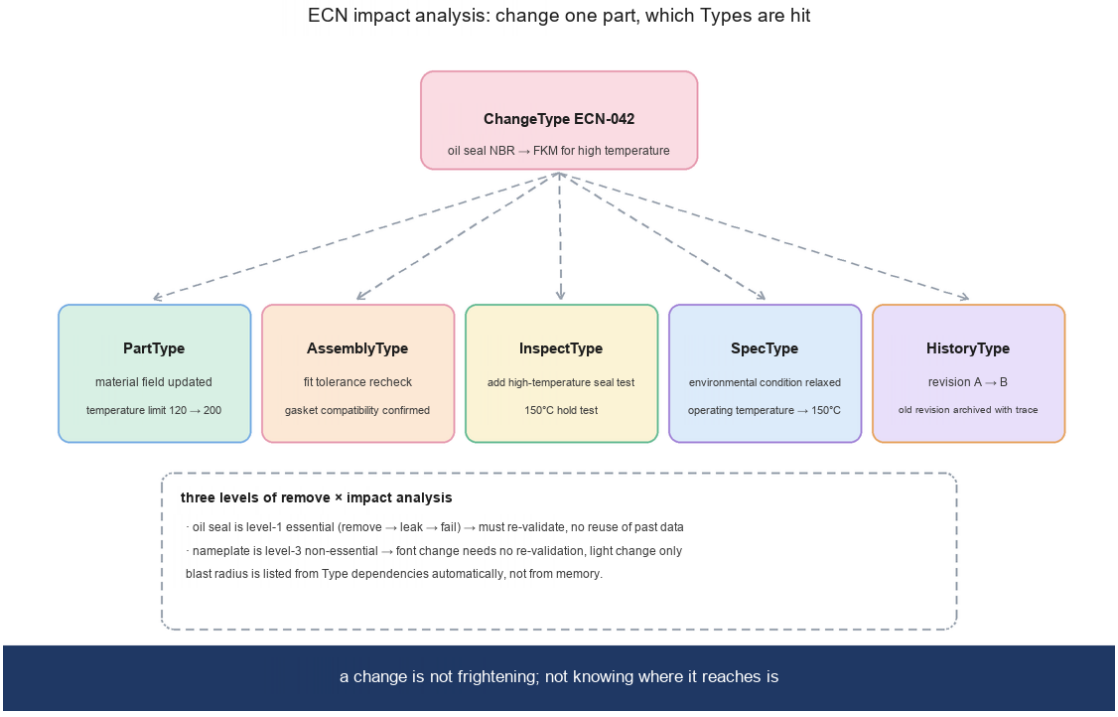


Figure 5-2 ECN impact analysis: three levels of remove × dependency chain, the blast radius listed automatically

Table 5-7 S6 output - engineering change notice ECN-042 (ChangeType)

Field	Content
cause	the customer's environment has become harsher and the upper temperature limit must rise from 120°C to 150°C
change content	output shaft oil seal material NBR → FKM (PartType PT-SF-006 revision)

Field	Content
impact analysis	PartType: PT-SF-006 material field; AssemblyType: AS-HS-01 gasket compatibility recheck; InspectType: add a 150°C hold test; SpecType: revise the SPEC §6 environmental clause
remove judgement	The oil seal is a level-1 essential part (remove it and it leaks, then fails) → mandatory re-validation, previous case data may not be reused
re-validation items	high-temperature seal hold test, lip wear re-measurement, 8 h continuous run-in with temperature-rise re-measurement
effective revision	Rev B (Rev A drawing stamped void and filed)

Table 5-7 The impact-analysis column is derived automatically from the AssemblyType dependency chain; the engineer confirms rather than remembers.

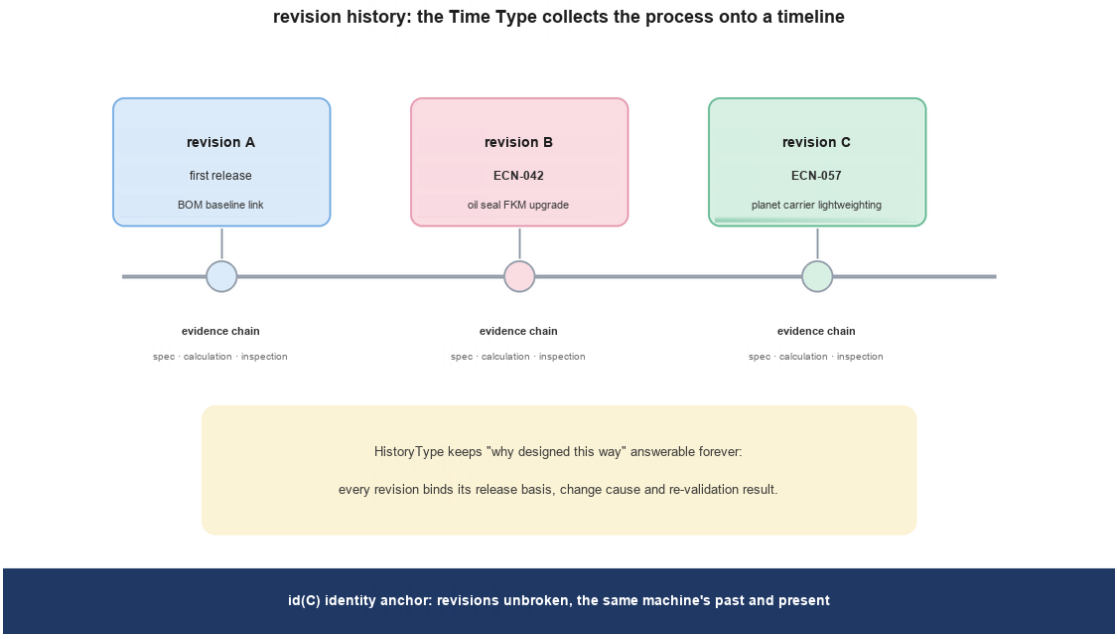


Figure 5-3 Revision history: id(C) keeps revisions unbroken, each with its full evidence attached

Table 5-8 S7 output - revision history (HistoryType excerpt)

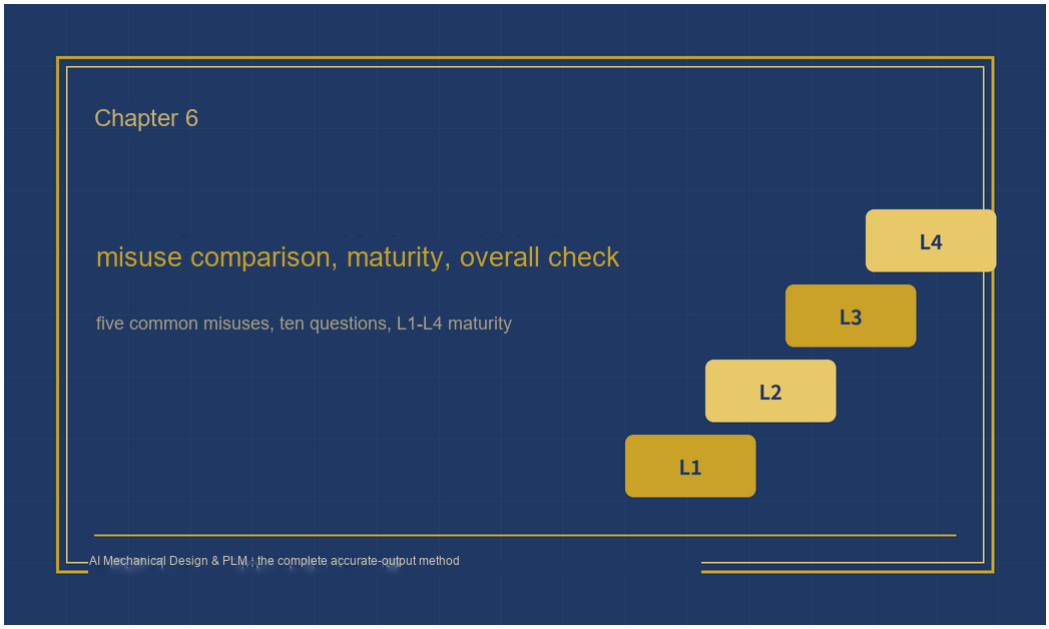
revision	date	bound change	release basis	Sign-off
Rev A	2026-03-12	—	type tests all passed	design / QA / manufacturing
Rev B	2026-06-01	ECN-042 Oil Seal FKM	150°C test passed	design / QA
Rev C	2026-09-20	ECN-057 Planet Carrier Lightweighting	fatigue test 200 h passed	design / QA / manufacturing

Table 5-8 Feedback to the Type Library: this case deposited 9 PartTypes and fixed 3 moves (design search, causal reasoning, refactoring).

Chapter focus

Each S1-S7 stage does one thing only, produces one Type, and answers to one set of criteria.
The soul of a prompt is its acceptance clauses: no fabrication, nothing written without a source, vague judgements are out.
The full planetary gearbox example shows the six Types as real entities: one line from SPEC-PG-001 to Rev C.
Impact analysis is computed from the dependency chain and the three levels of remove, not from memory.

Chapter 6 Misuse Comparison, Maturity, Overall Check



With the method taught, the last chapter handles the close: first the five most common misuses - between them they cover about eighty percent of real-world failure; then the ten-question checklist to score your own case; finally the maturity model to plan the team's next step.

6.1 Five Misuses

The five misuses share one root: Type decoupled from number, output decoupled from criteria. None of them blows up immediately, but all of them quietly leak "accurate output".

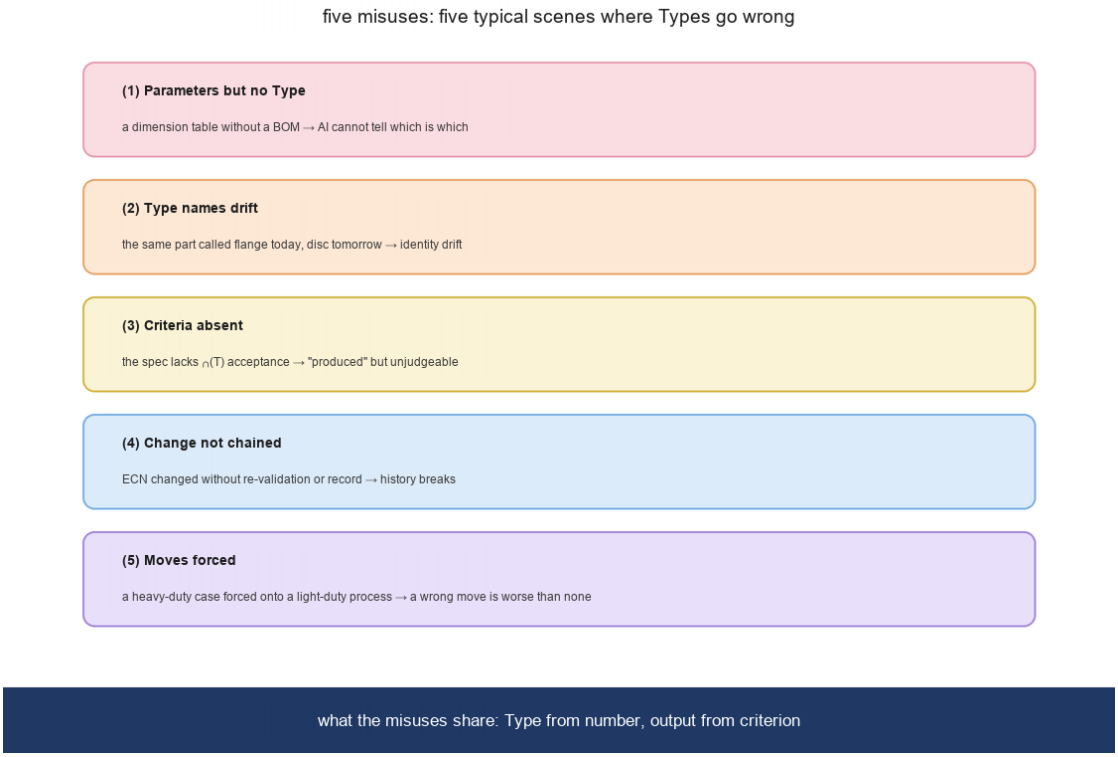


Figure 6-1 Five misuses: five typical scenes where Type and number come apart

Table 6-1 Misuse comparison table

Misuse	Symptom	Consequence	Correction
(1) Parameters without a Type	throwing over only a dimension table	AI cannot tell one from another	establish the PartType and AssemblyType before filling in numbers
(2) Type names drift	one part, many names; renamed but not renumbered	identity drifts, BOM does not match	id(C) anchors the number; renames leave a trace
(3) Criteria absent	the specification has no acceptance condition	produced but impossible to judge	∩(T) quantified, three-valued judgement
(4) Changes not linked to the chain	ECN changed without re-validation	the history breaks and the field fails	dependency chain + three levels of remove must be listed for re-validation
(5) Moves forced onto the wrong situation	a heavy-load case run through a light-load process	a misapplied move is worse than none	choose a move by situation; when stuck, switch

Table 6-1 Pin this table on the project-room wall: five sentences, five alarm bells.

6.2 The Ten-Question Checklist

Ask yourself ten questions before closing a case. Each must be answered with a Type, never with "probably", "should be" or "as I recall".

Table 6-2 The ten closing questions

#	Problem	what a passing answer looks like
1	Does every part have a PartType?	list every item by number, none missed
2	Has the BOM criterion been applied?	every item is tagged with its remove level
3	Are the container boundaries drawn clearly?	each housing / module has its id(C) and boundary description
4	Is every specification quantified?	∩(T) clauses are measurable and decidable
5	Can every number state its source?	No blank [Source] fields
6	Is the derivation complete?	known → formula → substitution → result, every step checkable
7	Has engineering validation been done?	tolerance chain, strength and performance each have a report
8	Is every change tagged with an impact analysis?	Every ECN carries a dependency-chain list
9	Can the history answer "why"?	any revision's release basis is traceable
10	Has the Type Library accumulated?	the experience of this case has been fed back to the Type Library

Table 6-2 Only when all ten questions pass is the output accurate; if one is stuck, go back to the corresponding step and redo it.

6.3 Maturity L1-L4

Accuracy is not a switch but a staircase. A team can assess which level it stands on, then decide where to climb.

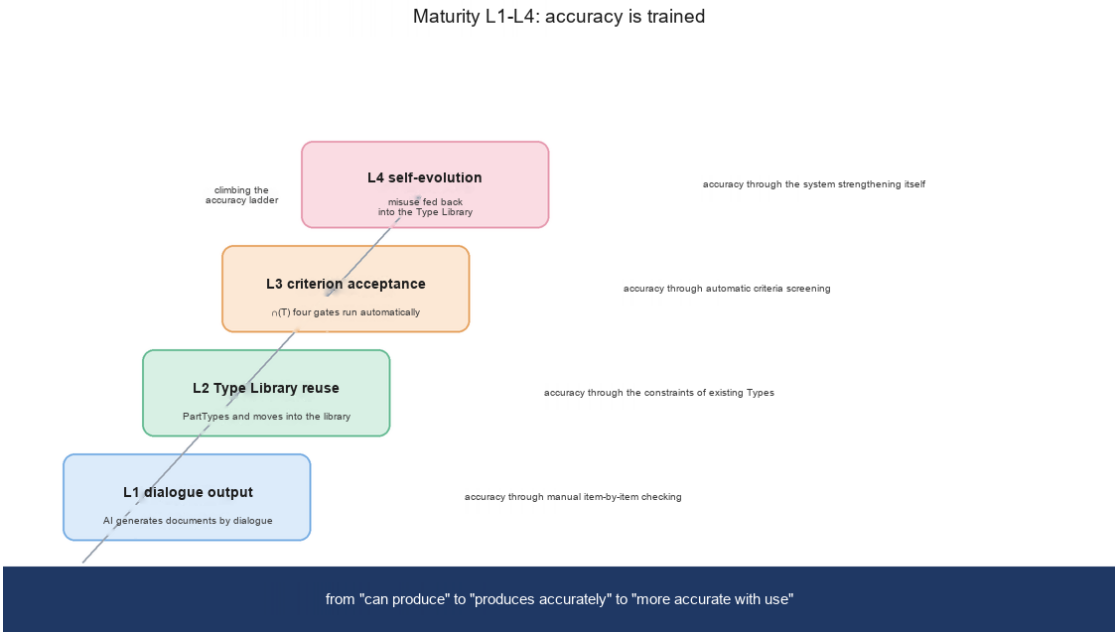


Figure 6-2 Maturity L1-L4: from being able to produce, to producing accurately, to getting more accurate with use

Table 6-3 Maturity model

Grade	Feature	what accuracy rests on	Upgrade action
L1 dialogue output	AI generates documents through dialogue	manual item-by-item check	build the field templates of the Type
L2 Type Library reuse	PartTypes and moves enter the library	constraints of existing Types	write an ∩(T) criterion for every Type
L3 criterion acceptance	the four-gate funnel runs automatically	criteria screen automatically	link the Type evidence chain and the history
L4 Self-evolution	misuses fed back to the Type Library	the system strengthens itself	periodically refactor moves and criteria

Table 6-3 Most teams start at L1; reaching L2 within three months and touching L3 within a year is a realistic pace.

three closing questions for the review

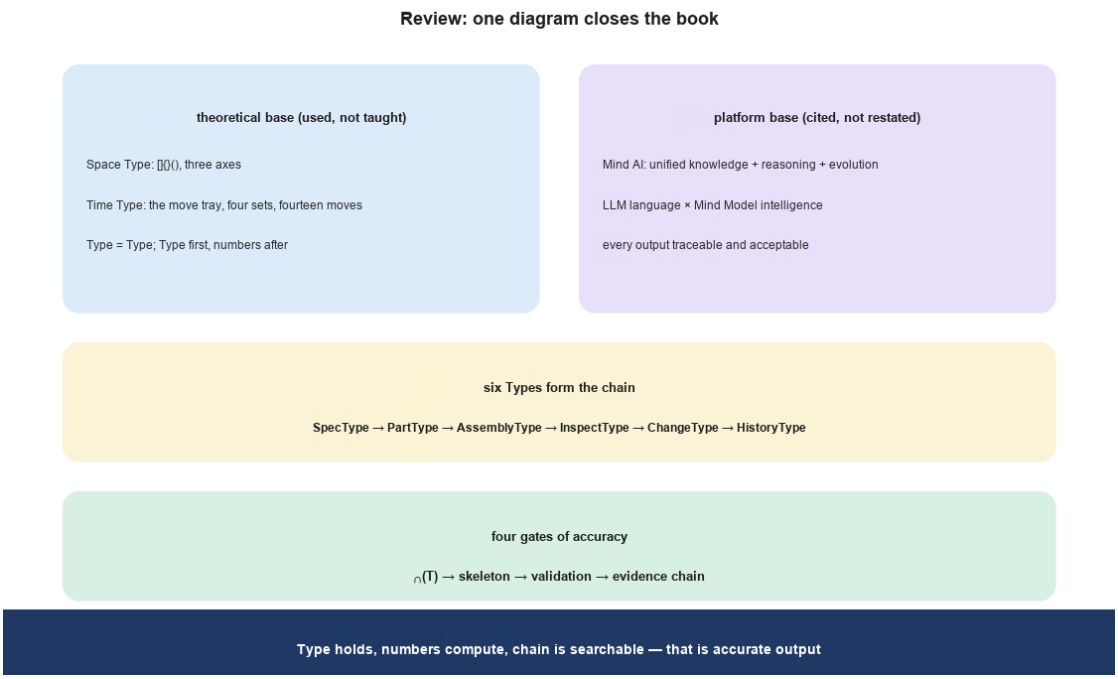


Figure R-1 One diagram to close: theory as tool, platform as base, six Types as a chain, four gates guarding accuracy

After six chapters, everything closes into three questions for each reader to take back to their own floor.

Question one - have the Types been established?

In your own case, is every part, every assembly and every specification a Type - or just a pile of files? First the Type, then the number: that is this book's first habit and the precondition of all accuracy.

Question two - has accuracy been held?

∩(T) criteria, deductive skeleton, engineering validation, Type evidence chain - how many of the four gates does your process have? Add whichever is missing; you need not do it all at once, but you must know where the gap is.

Question three - will it be more accurate tomorrow?

Have this misuse, this ECN and this measurement surprise all been fed back to the Type Library? L4 self-evolution is not platform magic; it is the accumulation of that one step, "archiving and evolution", at the close of every case.

Table R-1 Overall checklist for the book

Aspect	Checkpoint	Pass standard
Output Type	all six Types (spec, part, assembly, inspection, change, history) complete	every Type has fields, criteria and neighbours
Criteria	hang the four ∩(T) gates into the process	Non-conforming output is stopped automatically
Derivation	the deductive skeleton is checkable	every step has units, a formula and a source
Verification	all three engineering validation routes complete	tolerance chain, strength and performance each have a report
traceability	the Type evidence chain is complete	any number's origin found within three minutes

Aspect	Checkpoint	Pass standard
Evolution	the Type Library keeps accumulating	every closed case has a feedback record

Table R-1 This table can serve as a quarterly audit sheet.

Appendix A Quick Reference: Terms, Moves, Notation

Table A-1 Terminology quick reference

Terminology	English	In one sentence
Type	Type	something with boundaries, composition, criteria and a name
property / property value	Property / Value	[ ] Axis: circle differences into dimensions and values
BOM	Bill of Materials	{ } Axis: composition structure and levels
Spatial Container	Space Container	( ) Axis: space with boundaries and identity
Intersection criterion	Intersection Criterion ∩(T)	the intersection of necessary conditions that must all hold before entering the Type
Identity anchor	Identity Anchor id(C)	an identity marker that keeps revisions unbroken
Three levels of remove	Remove Levels	change grading for essential / functional / non-essential parts
Type set	Type Set T = (F(T), ∩(T))	a complete notation of structure description plus criteria
Four framework steps	Framework Four Steps	Distinguish similarity and difference, trigger abstraction, name the dimension, judge membership
Three-layer judgement	Three-Layer Judgment	the judgement order container → composition → abstract
Move	Motion / Function	Time Type: a repeatable R&D method
Type evidence chain	Type Evidence Chain	a traceability chain of one source per number, one ledger per Type

Table A-1 The word "Type" is always rendered Type in English, never Form.

Table A-2 The fourteen moves of the Time Type × where they land in mechanical design and PLM

Set	Move	where it lands in mechanical R&D
Search and extraction	Type-time function search	find existing PartTypes and past cases in the Type Library
Search and extraction	Design search and extraction	catalogue selection, patent and benchmark comparison
Search and extraction	Design patterns	reuse mature configurations such as the planetary train and planet carrier
generative application	Logic generation and application	configuration combination generation under constraints
generative application	Function generation and application	calculation templates: strength, life, efficiency
generative application	design - creation and innovation	new transmission architecture, lightweight configuration
reasoning and thinking	Reasoning	causal (failure analysis), statistical (tolerance), reverse (back-solving)
reasoning and thinking	Thinking methods	first principles to decompose the requirement, level thinking to change the solution
reasoning and thinking	Analysis	requirement analysis, tolerance-chain analysis, market analysis
reasoning and thinking	Problem solving	locating field problems such as interference, abnormal noise and temperature rise
evolutionary exploration	Continuous learning	measurements and customer complaints feed back to the criteria library
evolutionary exploration	Refactoring and recombination	Rebuild the BOM and drawing baseline after the ECN
evolutionary exploration	R&D and exploration	exploring the feasibility of new materials and processes
evolutionary exploration	Self-evolution	misuse cases settle automatically into screening rules

Table A-2 Four sets, fourteen moves: search and extraction, generative application, reasoning and thinking, evolutionary exploration.

Table A-3 Notation quick reference

Notation	Reading path	Meaning
[ ]	Abstract Set	Property X axis: the intersection of features generates the category
{ }	Composition Set	BOM Y axis: composition structure
( )	Container Set	Space Z axis: boundaries and identity
T = (F(T), ∩(T))	Type set notation	F(T) structure description + ∩(T) criterion
∩(T)	Intersection criterion	the intersection of necessary conditions for entering the Type
id(C)	container identity	identity anchor: revisions never break
remove(p, M)	removal judgement	after removing p, is M still M
S1–S7	the seven-stage pipeline	the prompt sequence from requirement clarification to archiving and evolution

Table A-3 The notation follows the original theoretical work; this book only uses it, it does not teach it.

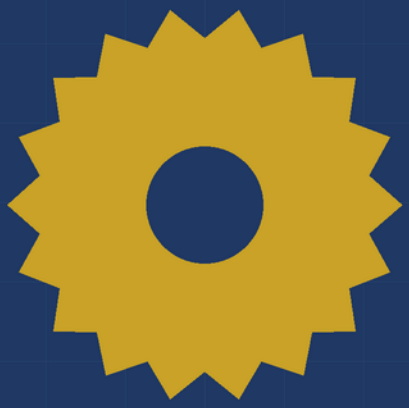
end of the book

The Type holds up, the numbers compute correctly, the chain can be searched - that is accurate output.

Take the six Types back to your own floor, fit the four gates into your process, and write the misuses into the Type Library.

The next machine will be more accurate than this one.

ONTYPOLOGY x MIND AI



same or different · apart or together · Type and name

where every design output starts

the Type comes first numbers only then

identify the structure first parameters only then

establish the criterion first accuracy only then

Space Type gives the product a skeleton AI can reason about

Time Type gives R&D a set of repeatable moves

Mind AI gives every accurate output a traceable evidence chain